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Nishio

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(54) **CIRCUIT BOARD AND METHOD OF MANUFACTURING CIRCUIT BOARD**

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(2013.01); **H05K 1/115** (2013.01); **H05K**
3/4652 (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

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Primary Examiner — Timothy J Thompson

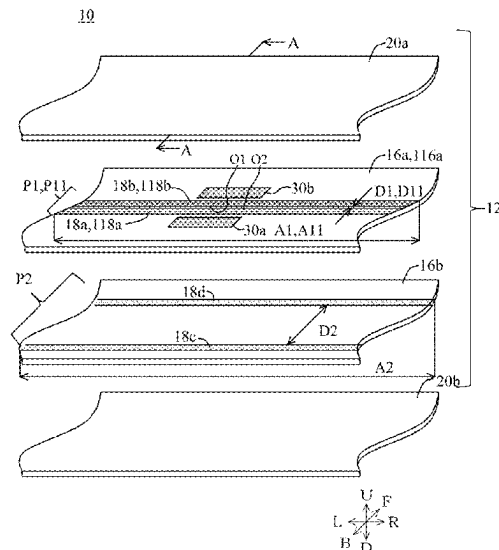
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(57) **ABSTRACT**

A circuit board includes one or more insulator layers, and upper and lower principal surfaces, and conductor layers in or on the substrate body. The conductor layers include a pair of first and second conductor layers on one of the insulator layers. Each of the pairs of the first and second conductor layers includes a first proximity section in which the first and second conductor layers are aligned in an orthogonal direction to an extending direction of the first conductor layer. A distance between the first and second conductor layers in the first proximity section is defined as a proximity distance. Most proximate first and second conductor layers are defined as the first and second conductor layers with a smallest proximity distance and are located on an upper principal surface of a first insulator layer of the one or more insulating layers.

20 Claims, 21 Drawing Sheets



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H05K 3/46 (2006.01)

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Fig.1

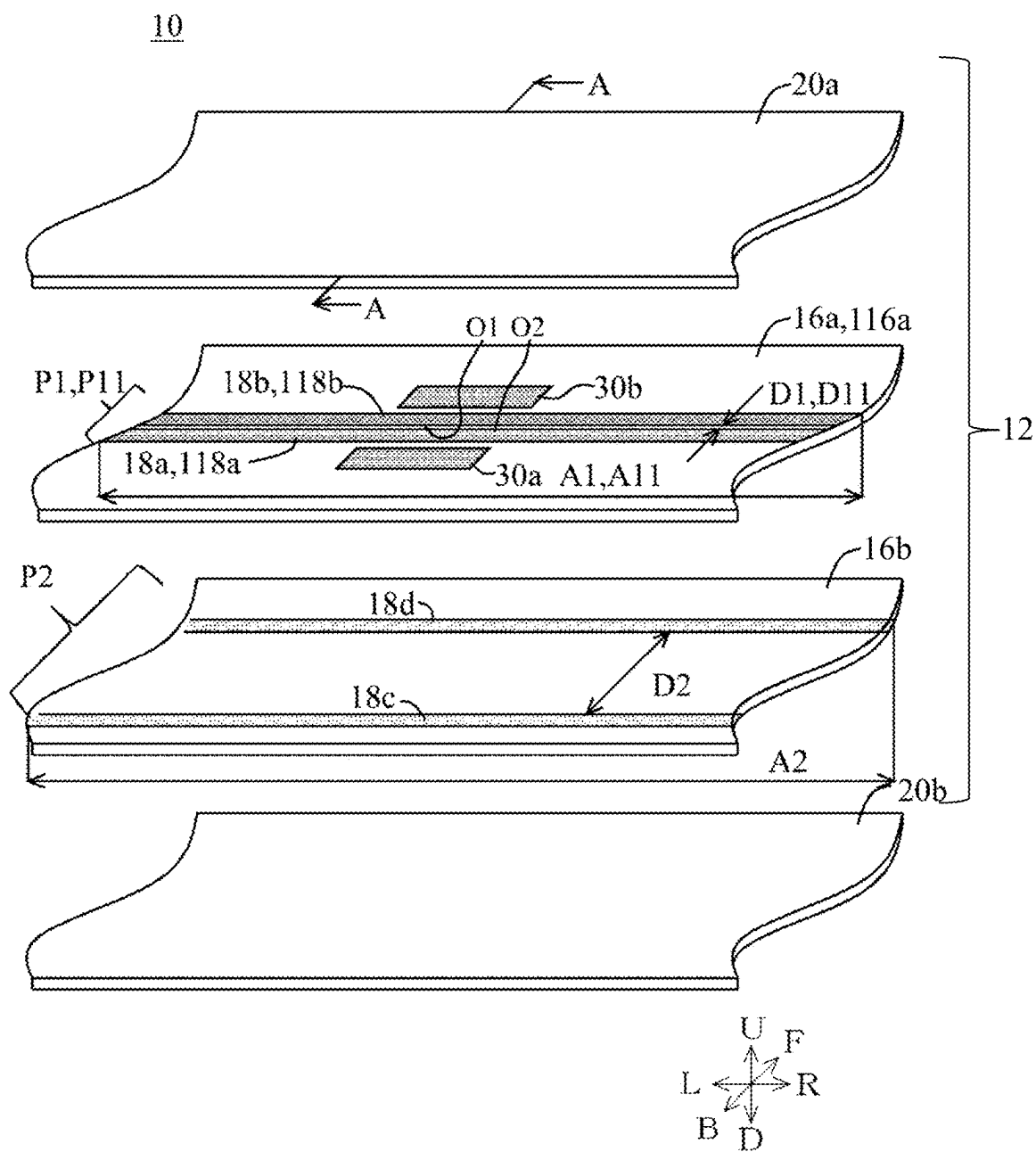


Fig.2

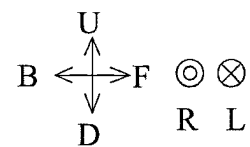
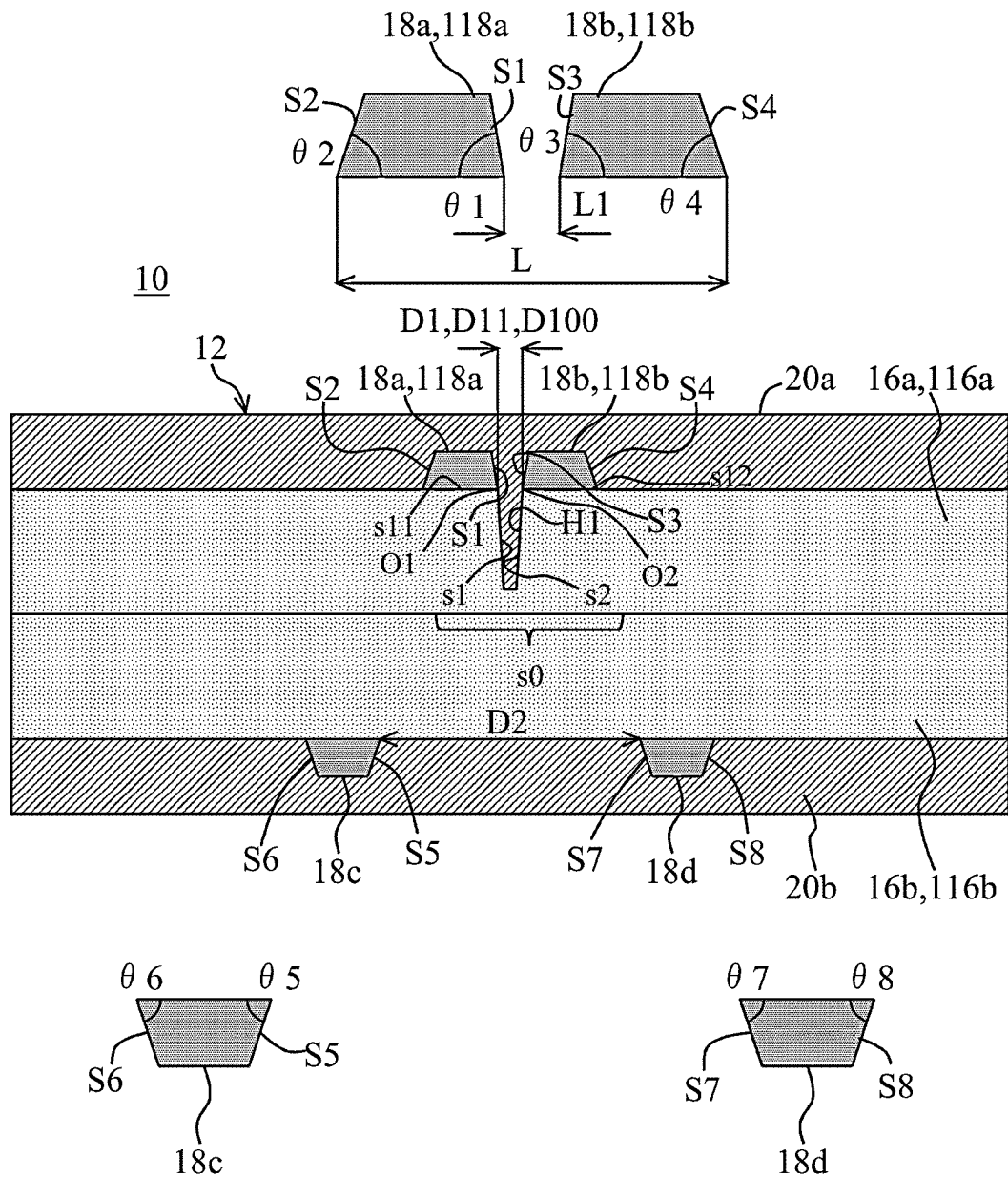


Fig.3

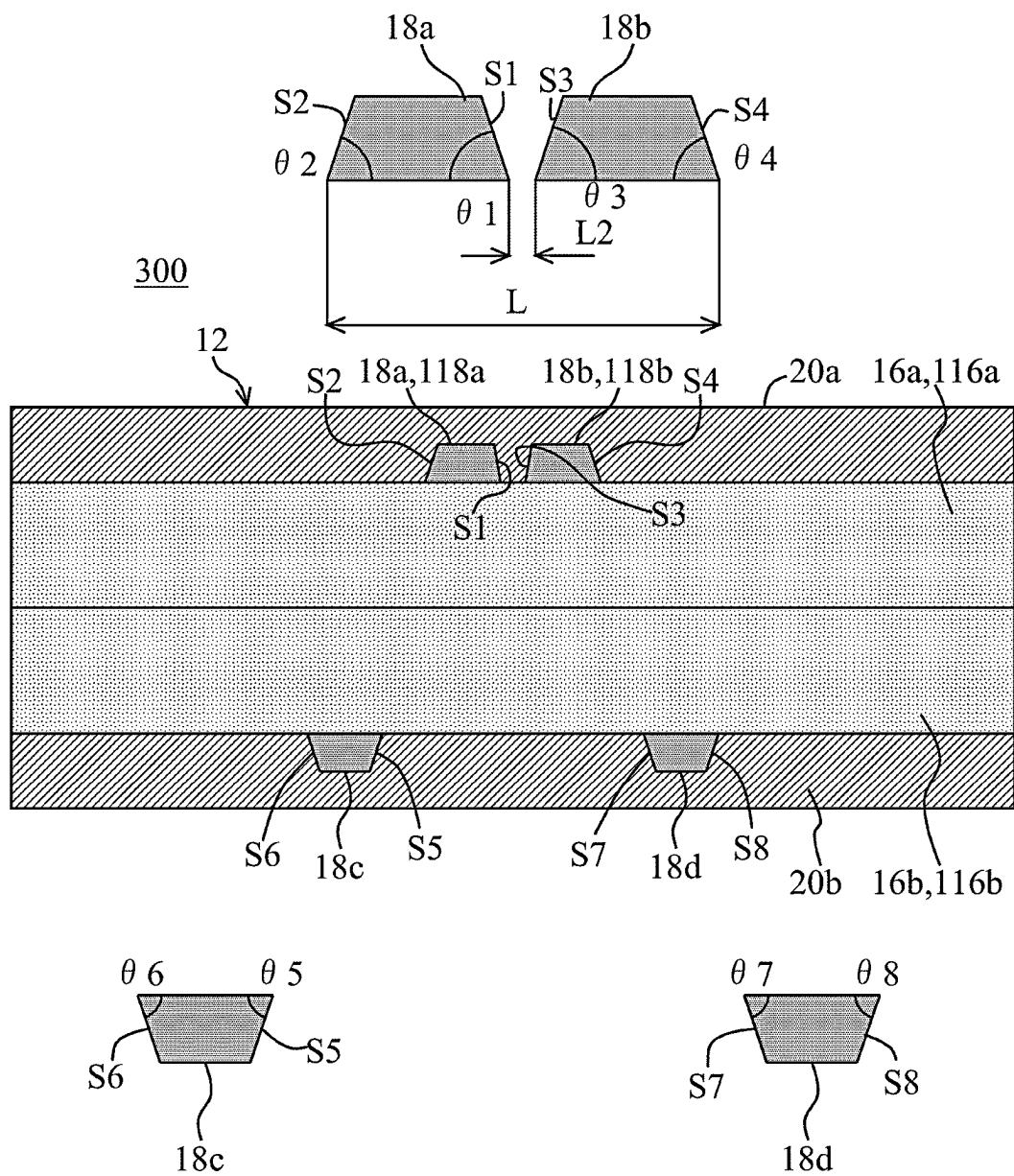


Fig.4

10a

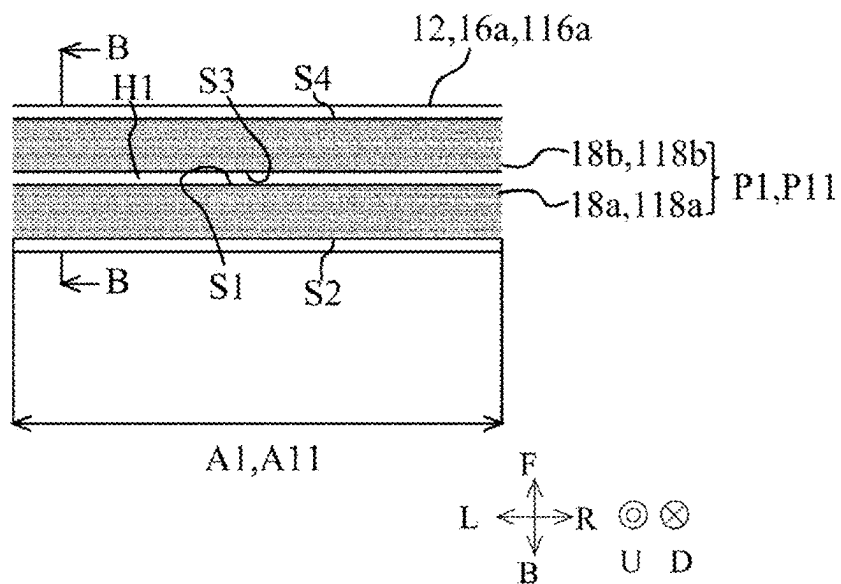


Fig.5

10a

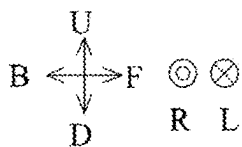
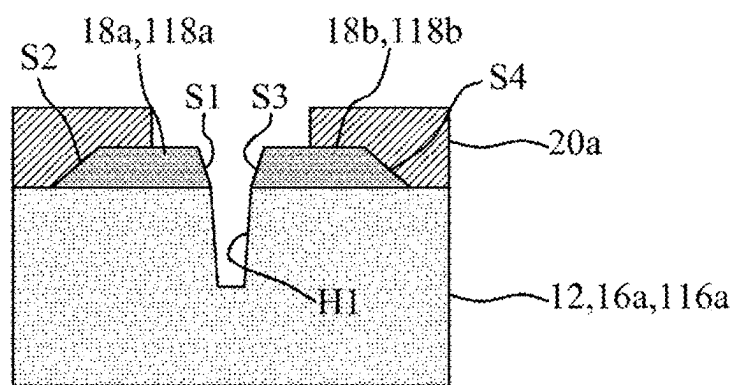


Fig.6

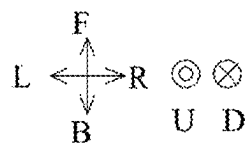
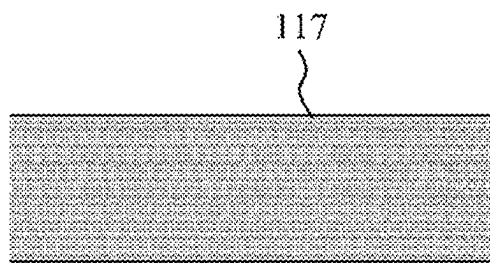


Fig.7

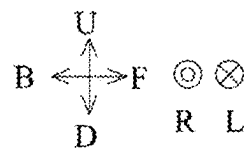
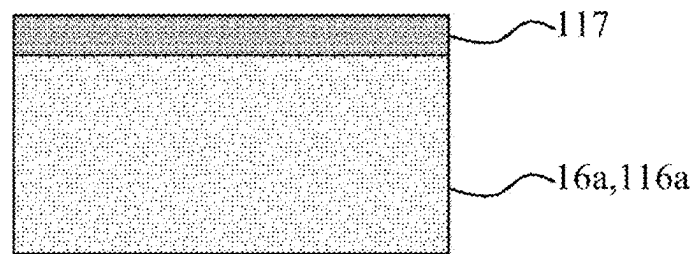


Fig.8

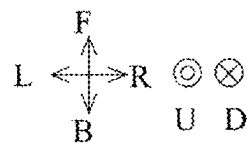
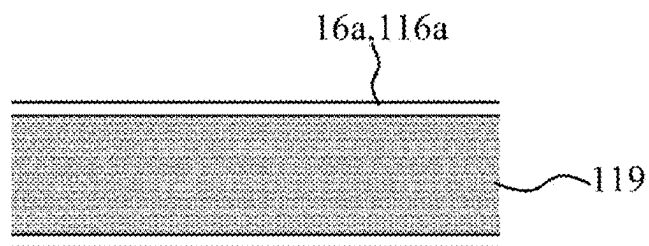


Fig.9

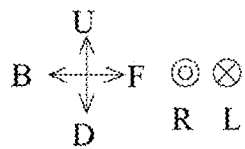
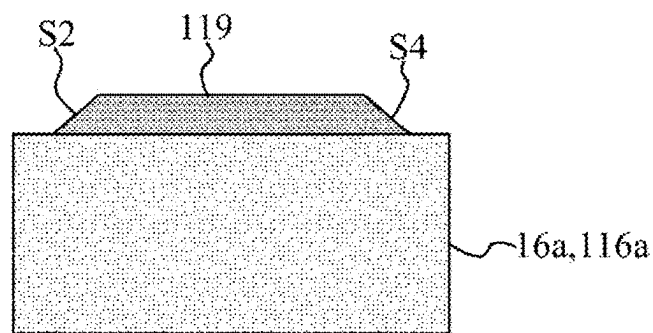


Fig.10

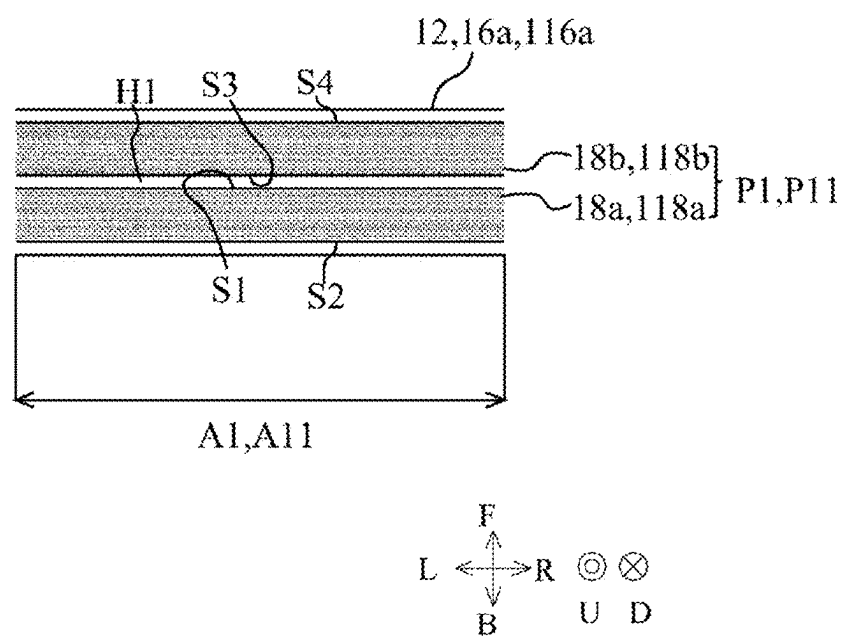


Fig. 11

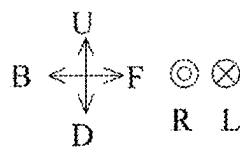
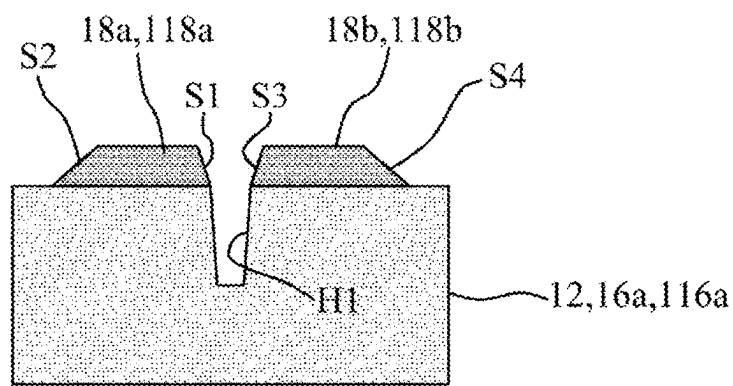


Fig.12

10b

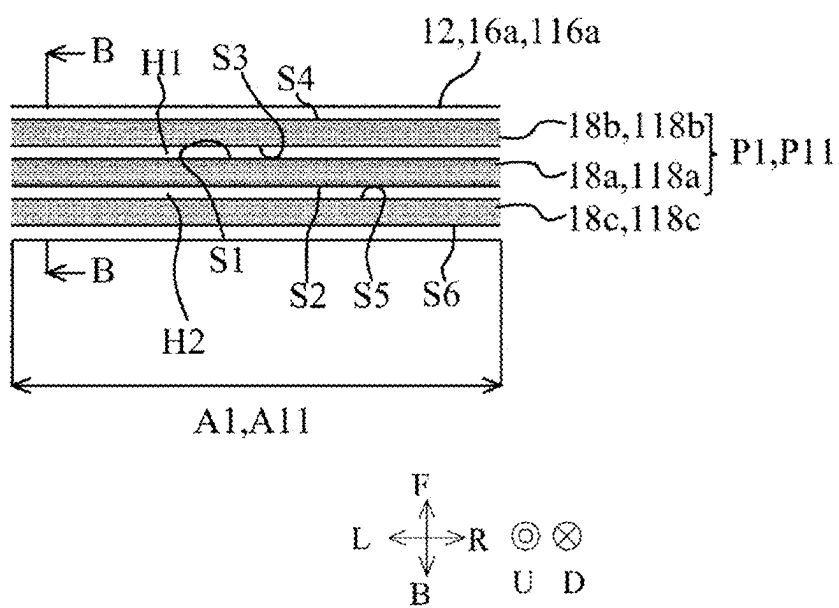


Fig.13

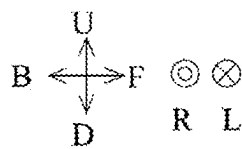
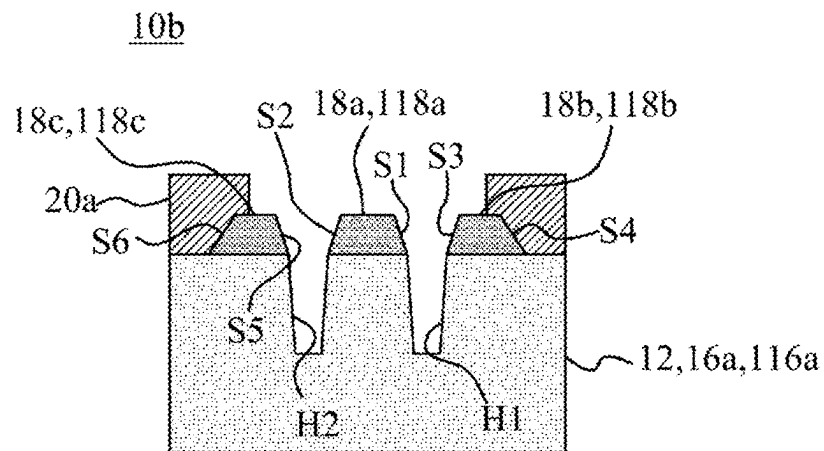


Fig.14

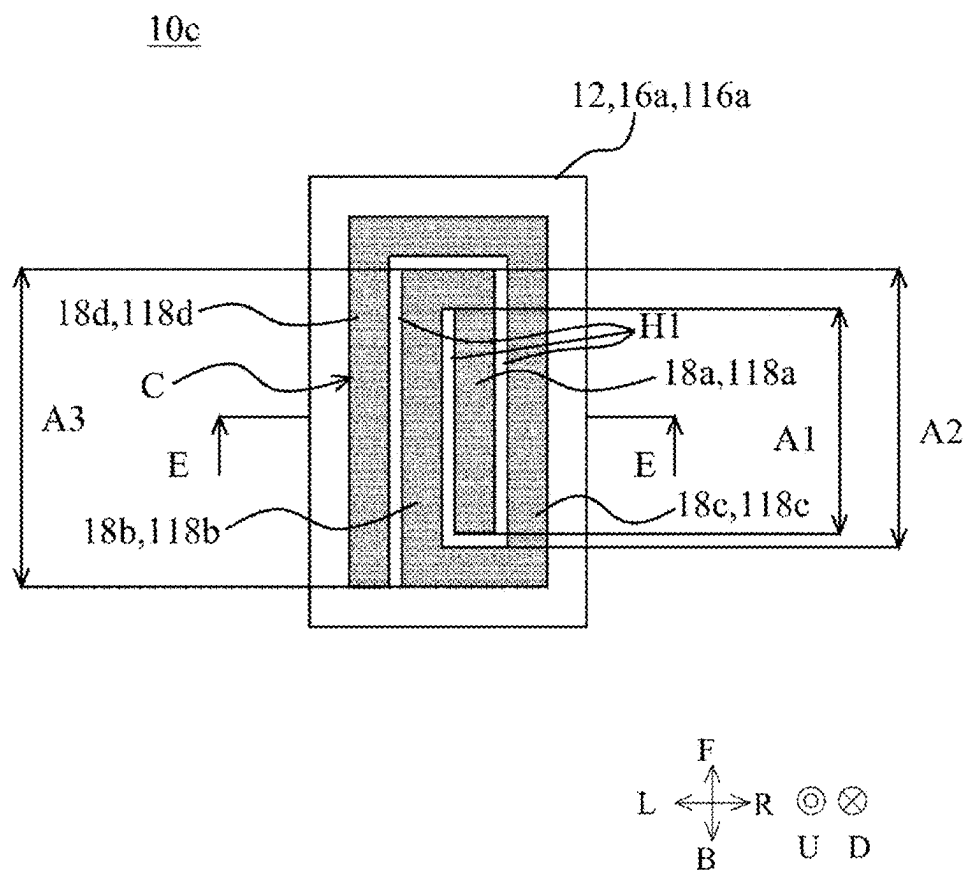
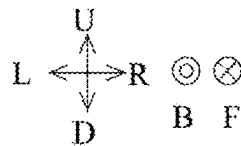
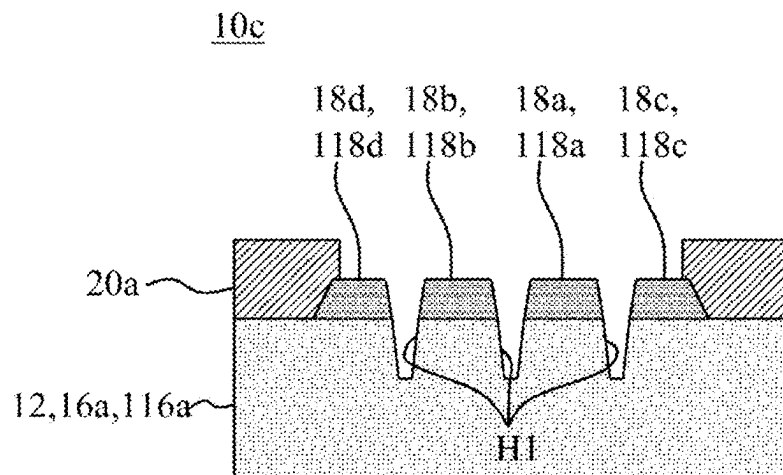


Fig.15



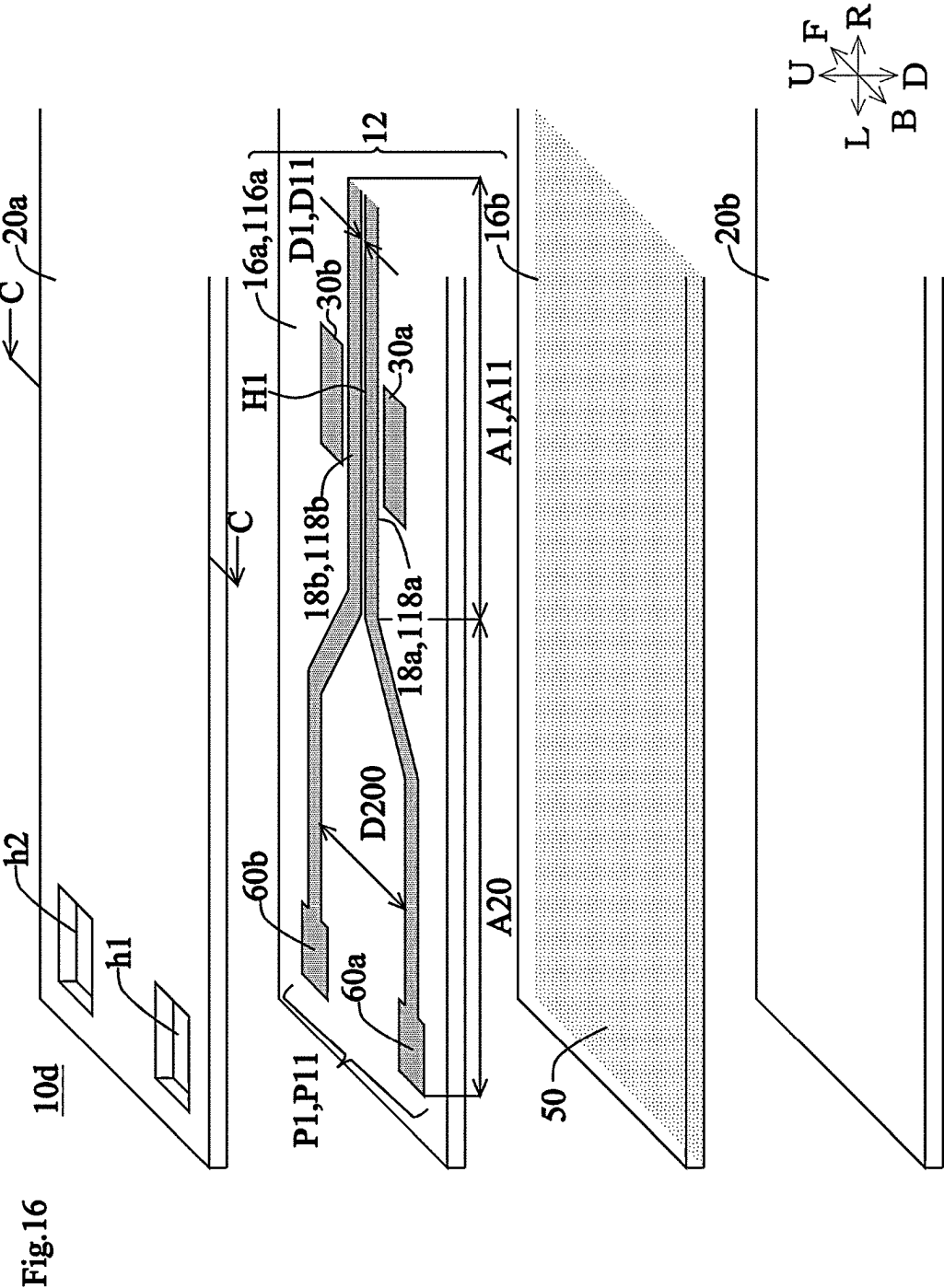


Fig.17

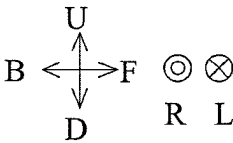
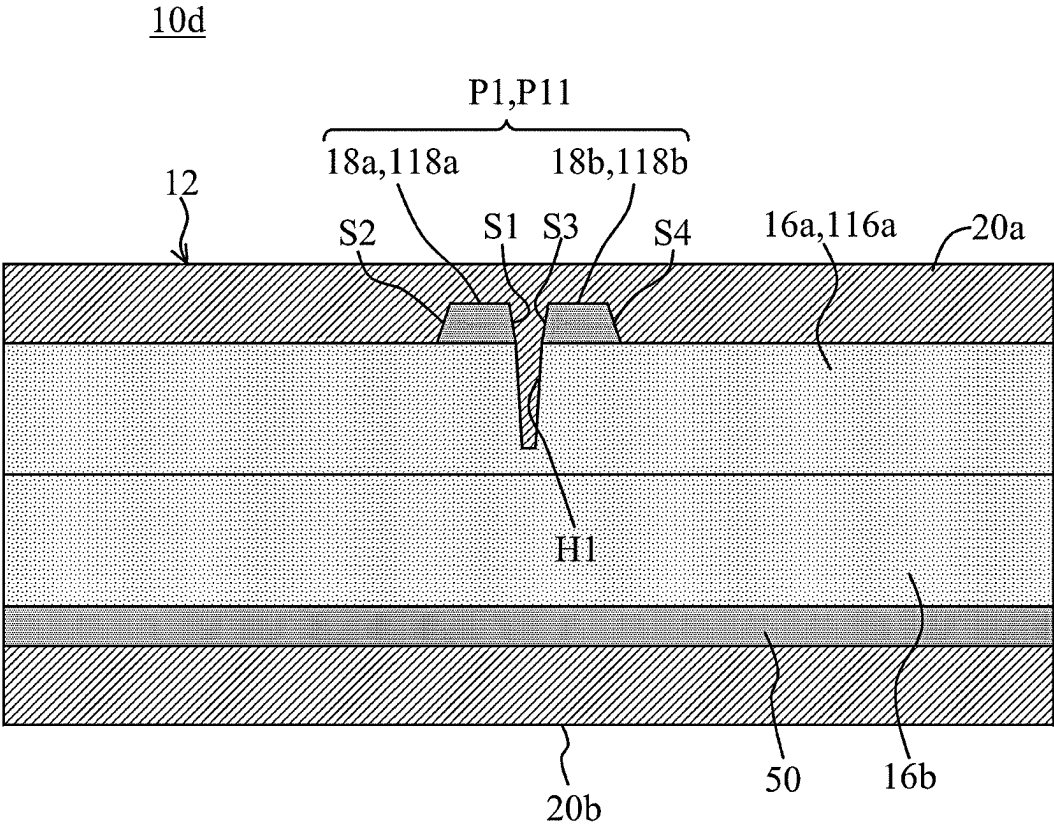


Fig. 19

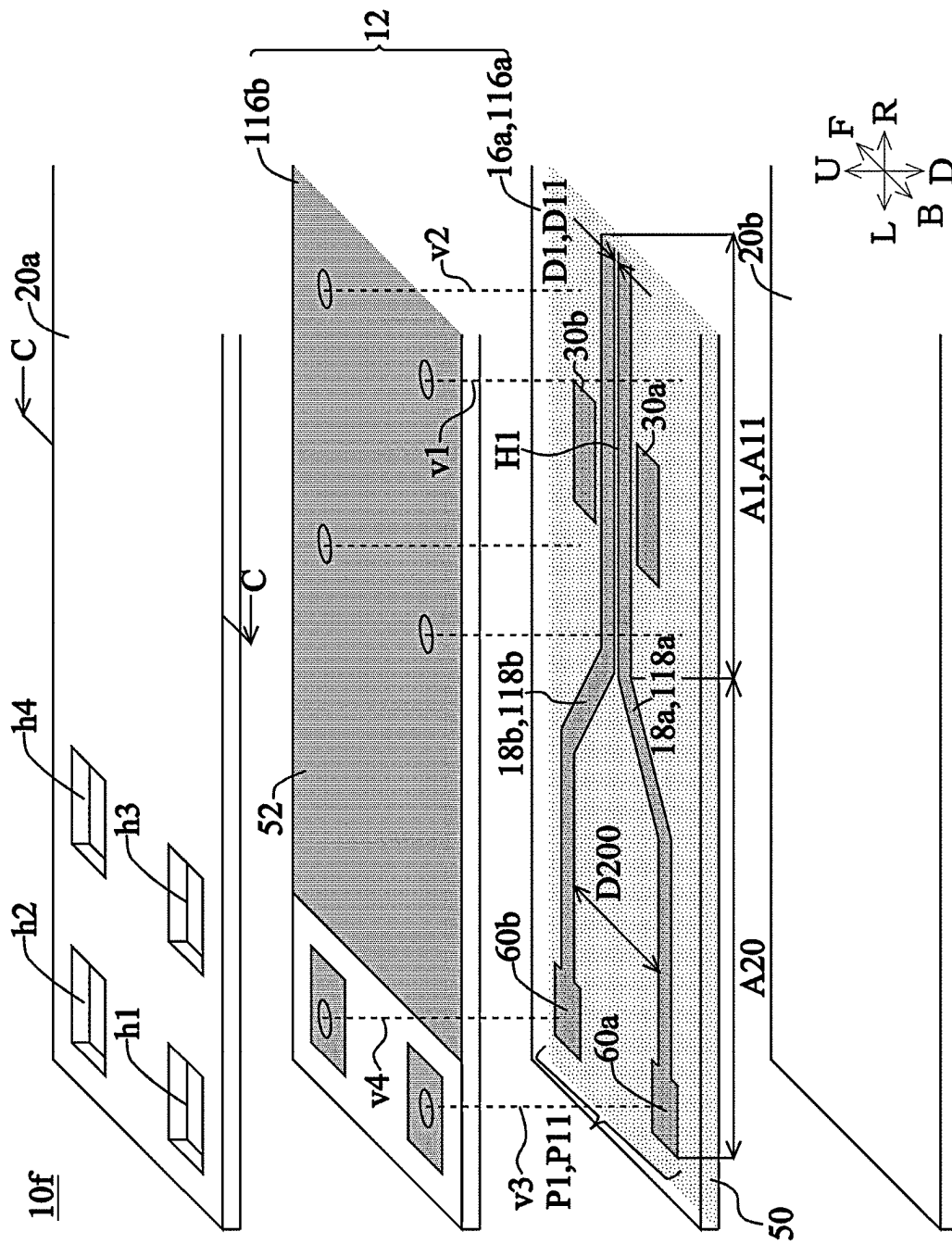


Fig.20

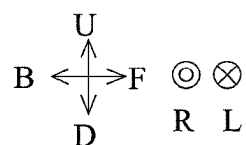
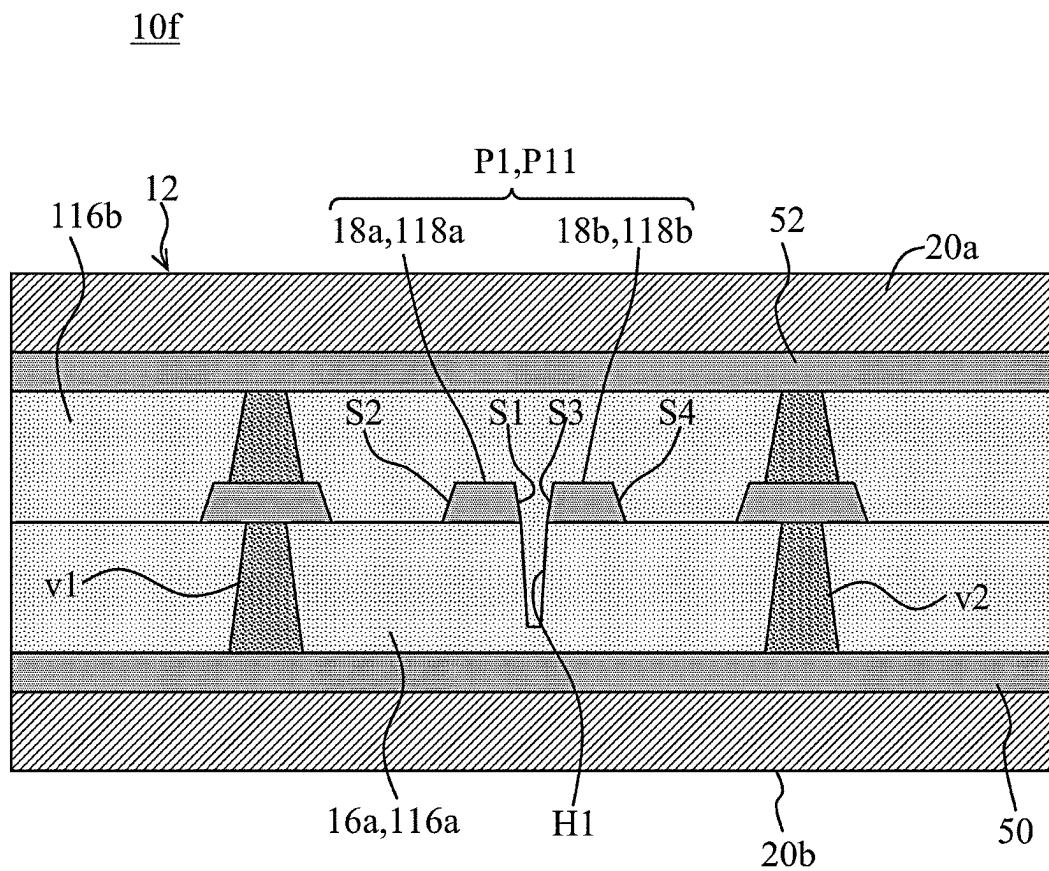
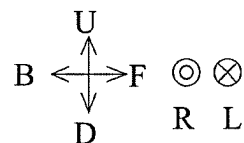
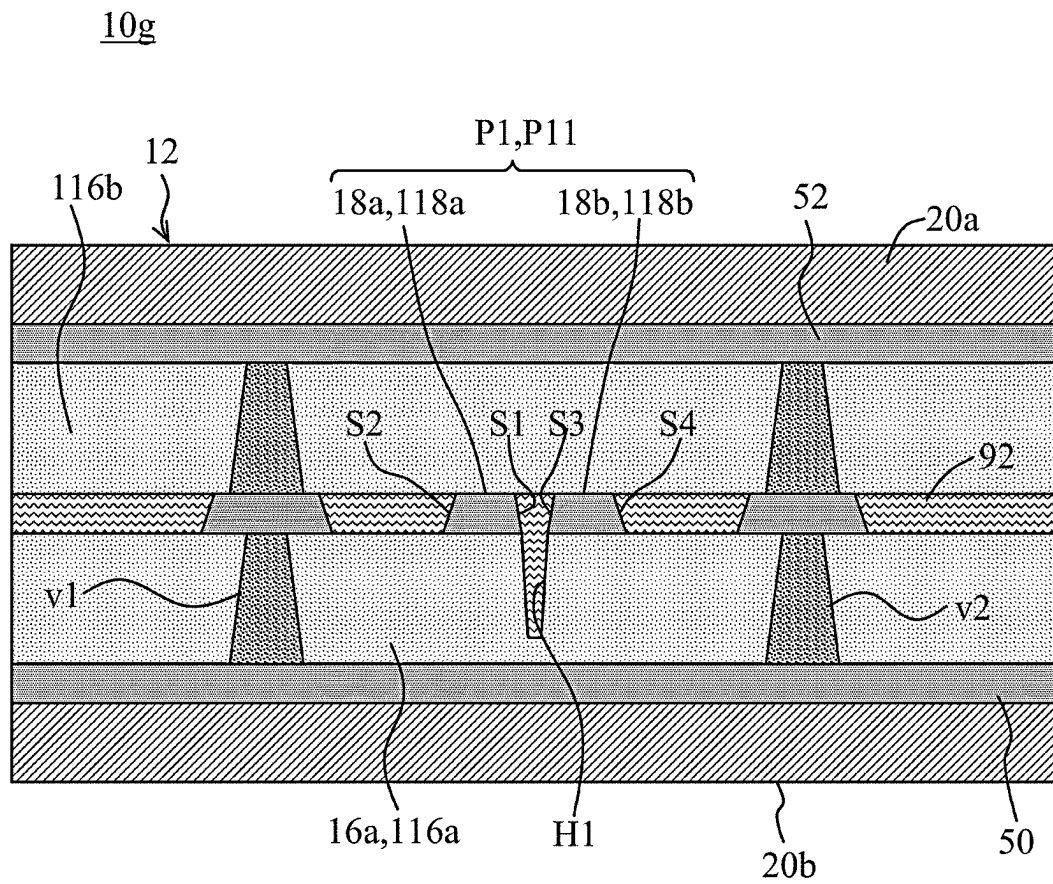


Fig.21



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**CIRCUIT BOARD AND METHOD OF
MANUFACTURING CIRCUIT BOARD****CROSS REFERENCE TO RELATED
APPLICATIONS**

This application claims the benefit of priority to Japanese Patent Application No. 2020-202486 filed on Dec. 7, 2020 and is a Continuation application of PCT Application No. PCT/JP2021/044203 filed on Dec. 2, 2021. The entire contents of each application are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION**1. Field of the Invention**

The present invention relates to a circuit board including a substrate body for a circuit including one or more insulator layers.

2. Description of the Related Art

Hitherto, as an invention related to a circuit board, for example, a composite transmission line described in Japanese Patent No. 6048633 is known. This composite transmission line includes a laminated insulator and multiple conductor patterns. The laminated insulator has a structure in which multiple insulator layers are stacked. The multiple conductor patterns are provided on the multiple insulator layers. The multiple conductor patterns form an electric circuit. The multiple conductor patterns are formed, for example, by processing metal foils provided on the multiple insulator layers by photolithography.

In the composite transmission line described in Japanese Patent No. 6048633, multiple conductor patterns provided on an insulator layer may be close to each other. In such a case, high accuracy is required to form the multiple conductor patterns. However, formation of multiple conductor patterns by photolithography has a limit in improving the accuracy of forming the multiple conductor patterns.

SUMMARY OF THE INVENTION

Preferred embodiments of the present invention provide circuit boards that each include a first conductor and a second conductor, which are close to each other and are formed with high accuracy, and methods of manufacturing such circuit boards.

A circuit board according to a preferred embodiment of the present invention includes a substrate body including one or more insulator layers, an upper principal surface, and a lower principal surface with a normal line extending in an up-down direction, and a plurality of conductor layers in or on the substrate body. The plurality of conductor layers include one or more pairs of a first conductor layer and a second conductor layer provided on one of the one or more insulator layers. Each of the one or more pairs of the first conductor layer and the second conductor layer includes a first proximity section in which the first conductor layer and the second conductor layer are aligned in an orthogonal direction orthogonal or substantially orthogonal to an extending direction of the first conductor layer. A distance between the first conductor layer and the second conductor layer in the first proximity section is defined as a proximity distance. The one or more insulator layers includes a first insulator layer. A most proximate first conductor layer and a

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most proximate second conductor layer are defined as the first and second conductor layers with a smallest proximity distance and are located on an upper principal surface of the first insulator layer. The most proximate first conductor layer includes a first side surface and a second side surface aligned in the orthogonal direction. The most proximate second conductor layer includes a third side surface and a fourth side surface aligned in the orthogonal direction. The first side surface and the third side surface face each other. A first acute angle between the first side surface and the upper principal surface of the first insulator layer is larger than a second acute angle between the second side surface and the upper principal surface of the first insulator layer.

A circuit board according to a preferred embodiment of the present invention includes a substrate body including one or more insulator layers, an upper principal surface, and a lower principal surface with a normal line extending in an up-down direction, and a plurality of conductor layers in or on the substrate body. The one or more insulator layers include a first insulator layer. The plurality of conductor layers include one or more pairs of a first conductor layer and a second conductor layer. The first conductor layer and the second conductor layer are provided on an upper principal surface of the first insulator layer. When viewed in the up-down direction, a first groove is provided in the upper principal surface of the first insulator layer in at least a portion of an area located between the first conductor layer and the second conductor layer. When viewed in the up-down direction, outer edges of the first groove overlap a first overlapping portion defined by a portion of an outer edge of the first conductor layer and a second overlapping portion being a portion of an outer edge of the second conductor layer. An inner surface of the first groove is located under the first overlapping portion and the second overlapping portion. A slope of the inner surface of the first groove changes, at the first overlapping portion, discontinuously downward with respect to a slope of a surface of the first insulator layer with which the first conductor layer is in contact. The inner surface of the first groove extends downward from the first overlapping portion.

A circuit board according to a preferred embodiment of the present invention includes a substrate body including one or more insulator layers, an upper principal surface, and a lower principal surface with a normal line extending in an up-down direction, and a plurality of conductor layers in or on the substrate body. The one or more insulator layers include a first insulator layer. The multiple conductor layers include one or more pairs of a first conductor layer and a second conductor layer. The first conductor layer and the second conductor layer are provided on an upper principal surface of the first insulator layer. When viewed in the up-down direction, a first groove is provided in the upper principal surface of the first insulator layer in at least a portion of an area located between the first conductor layer and the second conductor layer. A surface roughness of an inner surface of the first groove is different from a surface roughness of a surface of the first insulator layer with which the first conductor layer is in contact.

A method of manufacturing a circuit board according to a preferred embodiment of the present invention includes preparing an insulator layer with a metal foil in which a metal foil is provided on an upper principal surface of a first insulator layer including an upper principal surface and a lower principal surface aligned in an up-down direction, after the preparing step, forming a conductor layer from the metal foil by photolithography, and after the pattern forming step, linearly irradiating the conductor layer with a laser

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beam when viewed in the up-down direction to separate the conductor layer into a first conductor layer and a second conductor layer with an area irradiated with the laser beam interposed between the first conductor layer and the second conductor layer.

According to preferred embodiments of the present invention, the first conductor and the second conductor, which are close to each other, are able to be provided with high accuracy.

The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the preferred embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is an exploded perspective view of a circuit board 10 according to a preferred embodiment of the present invention.

FIG. 2 is a sectional view of the circuit board 10 taken along line A-A in FIG. 1.

FIG. 3 is a sectional view of a circuit board 300 of a comparative example.

FIG. 4 is a top view of a circuit board 10a according to a preferred embodiment of the present invention.

FIG. 5 is a sectional view of the circuit board 10a taken along line B-B.

FIG. 6 is a top view of the circuit board 10a during manufacturing.

FIG. 7 is a sectional view of the circuit board 10a during manufacturing.

FIG. 8 is a top view of the circuit board 10a during manufacturing.

FIG. 9 is a sectional view of the circuit board 10a during manufacturing.

FIG. 10 is a top view of the circuit board 10a during manufacturing.

FIG. 11 is a sectional view of the circuit board 10a during manufacturing.

FIG. 12 is a top view of a circuit board 10b according to a preferred embodiment of the present invention.

FIG. 13 is a sectional view of the circuit board 10b taken along line B-B.

FIG. 14 is a top view of a circuit board 10c according to a preferred embodiment of the present invention.

FIG. 15 is a sectional view of the circuit board 10c taken along line E-E.

FIG. 16 is an exploded perspective view of a circuit board 10d according to a preferred embodiment of the present invention.

FIG. 17 is a sectional view of the circuit board 10d taken along line C-C.

FIG. 18 is a sectional view of a circuit board 10e according to a preferred embodiment of the present invention.

FIG. 19 is an exploded perspective view of a circuit board 10f according to a preferred embodiment of the present invention.

FIG. 20 is a sectional view of the circuit board 10f taken along line C-C.

FIG. 21 is a sectional view of a circuit board 10g.

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DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Preferred Embodiment

Structure of Circuit Board 10

A structure of a circuit board 10 according to a preferred embodiment of the present invention will be described below with reference to the accompanying drawings. FIG. 1 is an exploded perspective view of the circuit board 10.

The circuit board 10 is, for example, a transmission line that transmits a first high-frequency signal to a fourth high-frequency signal. In the circuit board 10, a transmission direction of the first high-frequency signal to the fourth high-frequency signal is defined as a right-left direction. In other words, an extending direction of conductor layers 18a to 18d is defined as the right-left direction. A direction normal to an upper principal surface and a lower principal surface of a substrate body 12 of the circuit board 10 is defined as an up-down direction. The up-down direction and the right-left direction are orthogonal or substantially orthogonal to each other. An orthogonal direction orthogonal or substantially orthogonal to the up-down direction and the right-left direction is defined as a back-forth direction. The up-down direction, the right-left direction, and the back-forth direction of the circuit board 10 do not have to match the up-down direction, the right-left direction, and the back-forth direction of the circuit board 10 in use.

In the following, X is a component or member of the circuit board 10. In this specification, unless otherwise specified, portions of X are defined as follows. A front portion of X means a front half of X. A rear portion of X means a rear half of X. A left portion of X means a left half of X. A right portion of X means a right half of X. An upper portion of X means an upper half of X. A lower portion of X means a lower half of X. A front end of X means an end of X in a front direction. A rear end of X means an end of X in a rear direction. A left end of X means an end of X in a left direction. A right end of X means an end of X in a right direction. An upper end of X means an end of X in an upward direction. A lower end of X means an end of X in a downward direction. A front end portion of X means the front end of X and the vicinity thereof. A rear end portion of X means the rear end of X and the vicinity thereof. A left end portion of X means the left end of X and the vicinity thereof. A right end portion of X means the right end of X and the vicinity thereof. An upper end portion of X means the upper end of X and the vicinity thereof. A lower end portion of X means the lower end of X and the vicinity thereof.

The circuit board 10 includes the substrate body 12, the multiple conductor layers 18a to 18d, and protective layers 20a and 20b. The substrate body 12 has a plate shape. Consequently, the substrate body 12 includes an upper principal surface and a lower principal surface. The direction normal to the upper principal surface and the lower principal surface of the substrate body 12 extends in the up-down direction. The substrate body 12 extends in the right-left direction. The substrate body 12 has flexibility. Thus, the substrate body 12 is used in a bent state.

The substrate body 12 includes one or more insulator layers 16a and 16b. The substrate body 12 has a structure in which the insulator layers 16a and 16b are stacked in this order from top to bottom. The insulator layers 16a and 16b have the same shape as the substrate body 12 when viewed in the up-down direction. A material of the insulator layers 16a and 16b is, for example, a thermoplastic resin. The

thermoplastic resin is, for example, a liquid crystal polymer or polyimide. Thus, the insulator layers **16a** and **16b** have flexibility.

The conductor layers **18a** to **18d** are provided on the substrate body **12**. To be more specific, the conductor layer **18a** and the conductor layer **18b** are provided on an upper principal surface of the insulator layer **16a**. Consequently, the conductor layers **18a** to **18d** include a pair P1 of the conductor layer **18a** (first conductor layer) and the conductor layer **18b** (second conductor layer) provided on the insulator layer **16a** (first insulator layer). The conductor layer **18a** and the conductor layer **18b** have a linear shape extending in the right-left direction when viewed in the up-down direction. Thus, the pair P1 of the conductor layer **18a** (first conductor layer) and the conductor layer **18b** (second conductor layer) include a first proximity section A1 in which the conductor layer **18a** (first conductor layer) and the conductor layer **18b** (second conductor layer) are aligned in the back-forth direction (orthogonal direction orthogonal or substantially orthogonal to the extending direction of the first conductor layer). In the present preferred embodiment, the conductor layer **18b** is located in front of the conductor layer **18a**. Hereinafter, a distance between the conductor layer **18a** (first conductor layer) and the conductor layer **18b** (second conductor layer) in the first proximity section A1 is defined as a proximity distance D1. In the present specification, the proximity distance is an average of distances between two conductor layers in a proximity section. The conductor layer **18a** is a signal line that transmits the first high-frequency signal. The conductor layer **18b** is a signal line that transmits the second high-frequency signal. Thus, the conductor layer **18a** (first conductor layer) and the conductor layer **18b** (second conductor layer) are not electrically connected. In this specification, the high-frequency signal is, for example, a signal having a frequency of several GHz or higher.

The conductor layer **18c** and the conductor layer **18d** are provided on a lower principal surface of the insulator layer **16b**. The conductor layers **18a** to **18d** include a pair P2 of the conductor layer **18c** (first conductor layer) and a conductor layer **18d** (second conductor layer) provided on the insulator layer **16b**. The conductor layer **18c** and the conductor layer **18d** have a linear shape extending in the right-left direction when viewed in the up-down direction. The pair P2 of the conductor layer **18c** (first conductor layer) and the conductor layer **18d** (second conductor layer) includes a first proximity section A2 in which the conductor layer **18c** (first conductor layer) and the conductor layer **18d** (second conductor layer) are aligned in the back-forth direction (orthogonal direction orthogonal or substantially orthogonal to the extending direction of the first conductor layer). In the present preferred embodiment, the conductor layer **18d** is located in front of the conductor layer **18c**. Hereinafter, a distance between the conductor layer **18c** (first conductor layer) and the conductor layer **18d** (second conductor layer) in the first proximity section A2 is defined as a proximity distance D2. The conductor layer **18c** is a signal line that transmits the third high-frequency signal. The conductor layer **18d** is a signal line that transmits the fourth high-frequency signal. The conductor layer **18c** (first conductor layer) and the conductor layer **18d** (second conductor layer) are not electrically connected.

The proximity distance D1 is smaller than the proximity distance D2. Thus, the pair P1 of the conductor layer **18a** (first conductor layer) and the conductor layer **18b** (second conductor layer) having the smallest proximity distance is defined as a pair P11 of a most proximate first conductor layer **118a** and a most proximate second conductor layer

118b. The pair P11 of the most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** has a most proximate section A11 in which the most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** are aligned in the back-forth direction (orthogonal direction). The proximity distance D1 between the most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** in the most proximate section A11 (first proximity section A1) is defined as a first most proximate inter-conductor distance D11.

The insulator layer **16a** on which the most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** are provided is defined as a first insulator layer **116a**. The most proximate first conductor layer **118a** (first conductor layer) and the most proximate second conductor layer **118b** (second conductor layer) are provided on an upper principal surface of the first insulator layer **116a**. Thus, the insulator layers **16a** and **16b** include the first insulator layer **116a**.

Next, a sectional structure of the circuit board **10** will be described with reference to FIG. 2. FIG. 2 is a sectional view of the circuit board **10** taken along line A-A in FIG. 1. The most proximate first conductor layer **118a** has a first side surface S1 and a second side surface S2 aligned in the back-forth direction (orthogonal direction). The first side surface S1 is a front surface of the most proximate first conductor layer **118a**. The second side surface S2 is a rear surface of the most proximate first conductor layer **118a**. The most proximate second conductor layer **118b** has a third side surface S3 and a fourth side surface S4 aligned in the back-forth direction (orthogonal direction). The third side surface S3 is a rear surface of the most proximate second conductor layer **118b**. The fourth side surface S4 is a front surface of the most proximate second conductor layer **118b**. Consequently, the first side surface S1 and the third side surface S3 face each other.

A first acute angle $\theta 1$ between the first side surface S1 and the upper principal surface of the first insulator layer **116a** is larger than a second acute angle $\theta 2$ between the second side surface S2 and the upper principal surface of the first insulator layer **116a**. A third acute angle $\theta 3$ between the third side surface S3 and the upper principal surface of the first insulator layer **116a** is larger than a fourth acute angle $\theta 4$ between the fourth side surface S4 and the upper principal surface of the first insulator layer **116a**.

The conductor layer **18c** includes a fifth side surface S5 and a sixth side surface S6 aligned in the back-forth direction (orthogonal direction). The fifth side surface S5 is a front surface of the conductor layer **18c**. The sixth side surface S6 is a rear surface of the conductor layer **18c**. The conductor layer **18d** includes a seventh side surface S7 and an eighth side surface S8 aligned in the back-forth direction (orthogonal direction). The seventh side surface S7 is a rear surface of the conductor layer **18d**. The eighth side surface S8 is a front surface of the conductor layer **18d**. Consequently, the fifth side surface S5 and the seventh side surface S7 face each other.

A fifth acute angle $\theta 5$ between the fifth side surface S5 and the lower principal surface of the insulator layer **16b** is equal or substantially equal to a sixth acute angle $\theta 6$ between the sixth side surface S6 and the lower principal surface of the insulator layer **16b**. A seventh acute angle $\theta 7$ between the seventh side surface S7 and the lower principal surface of the insulator layer **16b** is equal or substantially equal to an eighth acute angle $\theta 8$ formed by the eighth side surface S8 and the lower principal surface of the insulator layer **16b**.

As illustrated in FIG. 2, when viewed in the up-down direction, in at least a portion of an area located between the conductor layer 18a (first conductor layer) and the conductor layer 18b (second conductor layer), a first groove H1 is provided in the upper principal surface of the first insulator layer 116a. To be more specific, when viewed in the up-down direction, in at least a portion of an area located between the most proximate first conductor layer 118a and the most proximate second conductor layer 118b in the most proximate section A11 (first proximity section A1) in the first insulator layer 116a, the first groove H1 is provided in the upper principal surface of the first insulator layer 116a. The first groove H1 extends along the most proximate first conductor layer 118a and the most proximate second conductor layer 118b over an entire or substantially an entire length of the most proximate section A11 when viewed in the up-down direction. Consequently, the first groove H1 is a groove extending in the right-left direction when viewed in the up-down direction.

The first groove H1 has a tapered shape in which a width in the back-forth direction increases from a bottom surface of the first groove H1 upward when viewed in the right-left direction. However, the first groove H1 is different from a recess provided by, for example, undulating the first insulator layer 116a in the up-down direction when thermocompression bonding the substrate body 12. Thus, the first groove H1 has a structure described below.

When viewed in the up-down direction, outer edges of the first groove H1 overlap a first overlapping portion O1, which is a portion of an outer edge of the most proximate first conductor layer 118a, and a second overlapping portion O2, which is a portion of an outer edge of the most proximate second conductor layer 118b. The outer edges of the first groove H1 are a rear side and a front side of the first groove H1 when viewed in the up-down direction. The outer edges of the first groove H1 are portions of an inner surface s0 of the first groove H1. The first overlapping portion O1 is a front side of the most proximate first conductor layer 118a when viewed in the up-down direction. The second overlapping portion O2 is a rear side of the most proximate second conductor layer 118b when viewed in the up-down direction. The inner surface s0 of the first groove H1 is located under the first overlapping portion O1 and the second overlapping portion O2. The inner surface s0 includes inner surfaces s1 and s2. The inner surface s1 is a rear portion of the inner surface s0 of the first groove H1. The inner surface s2 is a front portion of the inner surface s0 of the first groove H1. A slope of the inner surface s1 of the first groove H1 changes, at the first overlapping portion O1, discontinuously downward with respect to a slope of a surface s11 of the first insulator layer 116a with which the most proximate first conductor layer 118a is in contact. Thus, the inner surface s1 is bent at the first overlapping portion O1. The inner surface s1 of the first groove H1 extends downward from the first overlapping portion O1. A slope of the inner surface s2 of the first groove H1 changes, at the second overlapping portion O2, discontinuously downward with respect to a slope of a surface s12 of the first insulator layer 116a with which the most proximate second conductor layer 118b is in contact. Thus, the inner surface s2 is bent at the second overlapping portion O2. The inner surface s2 of the first groove H1 extends downward from the second overlapping portion O2. A slope of a surface is an angle between the surface and the up-down direction.

Here, determination of discontinuity in this specification will be described. First, the section in FIG. 2 is observed at a magnification of 100 times. At this time, an angle between

a surface extending the surface s11 of the first insulator layer 116a with which the most proximate first conductor layer 118a is in contact and a surface extending the inner surface s1 is measured. When this angle is, for example, about 30 degrees or less, the slope of the surface s11 and the slope of the inner surface s1 change continuously. When this angle is larger than about 30 degrees, the slope of the surface s11 and the slope of the inner surface s1 change discontinuously.

A surface roughness of the inner surface s1 of the first groove H1 is different from a surface roughness of the surface s11 of the first insulator layer 116a with which the most proximate first conductor layer 118a (conductor layer 18a) is in contact. Similarly, surface roughness of the inner surface s2 of the first groove H1 is different from surface roughness of the surface s12 of the first insulator layer 116a with which the most proximate second conductor layer 118b (conductor layer 18b) is in contact. Surface roughness is measured according to JIS B 0601:1994. A maximum height Ry is then used as the surface roughness. A lower limit of a reference length is, for example, about 10 μm . In the present specification, surface roughness of a first surface being different from surface roughness of a second surface is as follows. Ry1 is maximum surface roughness of the first surface. Ry2 is maximum surface roughness of the second surface. $Ry1 < Ry2$. At this time, if $Ry1 \times 0.1 \leq Ry2 - Ry1$ is satisfied, the surface roughness of the first surface is different from the surface roughness of the second surface.

A maximum value D100 of the width of the first groove H1 in the back-forth direction (orthogonal direction) is equal to or greater than the first most proximate inter-conductor distance D11. The width of the first groove H1 in the back-forth direction (orthogonal direction) has the maximum value D100 at an upper end of the first groove H1. In the present preferred embodiment, the maximum value D100 of the width of the first groove H1 in the back-forth direction is equal or substantially equal to the first most proximate inter-conductor distance D11. Consequently, the first side surface S1 and the inner surface of the first groove H1 are continuously connected. The third side surface S3 and the inner surface s0 of the first groove H1 are continuously connected. That is, the upper principal surface of the first insulator layer 116a is not visible between the first side surface S1 and the inner surface s0 of the first groove H1. The upper principal surface of the first insulator layer 116a is not visible between the third side surface S3 and the inner surface s0 of the first groove H1.

A lower end of the first groove H1 is located above a lower principal surface of the first insulator layer 116a. Consequently, a depth of the first groove H1 is smaller than a thickness of the first insulator layer 116a.

The protective layer 20a covers the upper principal surface of the substrate body 12. In the present preferred embodiment, the protective layer 20a covers the upper principal surface of the first insulator layer 116a. The protective layer 20a is, for example, a resist layer or a coverlay layer. A dielectric constant of the protective layer 20a is lower than a dielectric constant of the first insulator layer 116a. Thus, the first groove H1 is filled with an insulating material having a dielectric constant lower than that of the first insulator layer 116a.

The protective layer 20b covers the lower principal surface of the substrate body 12. In the present preferred embodiment, the protective layer 20b covers the lower principal surface of the insulator layer 116b. The protective layer 20b is, for example, a resist layer or a coverlay layer.

The conductor layers 18a to 18d as described above are formed by patterning a metal foil attached to the upper

principal surface of the insulator layer 16a or the lower principal surface of the insulator layer 16b. The metal foil is, for example, a copper foil. The patterning is performed by combining photolithography and laser beam irradiation, for example. To be specific, the second side surface S2, the fourth side surface S4, the fifth side surface S5, the sixth side surface S6, the seventh side surface S7, and the eighth side surface S8 are formed by, for example, photolithography. The first side surface S1 and the third side surface S3 are formed by, for example, laser beam irradiation. The first groove H1 is formed in the upper principal surface of the first insulator layer 116a (insulator layer 16a) by, for example, laser beam irradiation. Further, the laser beam irradiation makes the first acute angle $\theta 1$ and the third acute angle $\theta 3$ larger than the second acute angle $\theta 2$, the fourth acute angle $\theta 4$, the fifth acute angle $\theta 5$, the sixth acute angle $\theta 6$, the seventh acute angle $\theta 7$, and the eighth acute angle $\theta 8$.

Advantageous Effects

According to the circuit board 10, the most proximate first conductor layer 118a (conductor layer 18a) and the most proximate second conductor layer 118b (conductor layer 18b), which are close to each other, can be positioned with high accuracy. To be more specific, when viewed in the up-down direction, in at least a portion of the area located between the conductor layer 18a and the conductor layer 18b, the first groove H1 is provided in the upper principal surface of the first insulator layer 116a. That is, when viewed in the up-down direction, in at least a portion of the area located between the most proximate first conductor layer 118a and the most proximate second conductor layer 118b in the most proximate section A11 (first proximity section A1) in the first insulator layer 116a, the first groove H1 is provided in the upper principal surface of the first insulator layer 116a. Thus, on the circuit board 10, laser beam irradiation is performed to separate the most proximate first conductor layer 118a and the most proximate second conductor layer 118b. As a result, the most proximate first conductor layer 118a and the most proximate second conductor layer 118b can be separated from each other while the most proximate first conductor layer 118a and the most proximate second conductor layer 118b are in close proximity to each other. As described above, according to the circuit board 10, the most proximate first conductor layer 118a (conductor layer 18a) and the most proximate second conductor layer 118b (conductor layer 18b), which are close to each other, can be positioned with high accuracy.

The fact that the first groove H1 is formed by laser beam irradiation can be confirmed, for example, by the fact that the surface roughness of the inner surface s0 of the first groove H1 is different from the surface roughness of the surface s11 of the first insulator layer 116a with which the most proximate first conductor layer 118a is in contact.

According to the circuit board 10, the conductor layers 18a to 18d can be formed in a short time. For example, in forming the conductor layers 18a to 18d, laser beam irradiation may be performed without performing photolithography. However, in this case, all of the side surfaces of the conductor layers 18a to 18d need to be formed by laser beam irradiation. Thus, a time required to form the conductor layers 18a to 18d is longer. In the circuit board 10, the first acute angle $\theta 1$ and the third acute angle $\theta 3$ are larger than the second acute angle $\theta 2$, the fourth acute angle $\theta 4$, the fifth acute angle $\theta 5$, the sixth acute angle $\theta 6$, the seventh acute angle $\theta 7$, and the eighth acute angle $\theta 8$. That is, the second side surface S2, the fourth side surface S4, the fifth side

surface S5, the sixth side surface S6, the seventh side surface S7, and the eighth side surface S8 are formed by photolithography, for example. The first side surface S1 and the third side surface S3 are formed by, for example, laser beam irradiation. In the photolithography step, the second side surface S2, the fourth side surface S4, the fifth side surface S5, the sixth side surface S6, the seventh side surface S7, and the eighth side surface S8 can be simultaneously formed in a short time. Consequently, by combining laser beam irradiation and photolithography, the conductor layers 18a to 18d can be formed in a short time.

According to the circuit board 10, a decrease in strength of the insulator layers 16a and 16b can be reduced or prevented. To be more specific, for example, in forming the conductor layers 18a to 18d, the laser beam irradiation may be performed without performing photolithography. However, in this case, all of the side surfaces of the conductor layers 18a to 18d need to be formed by laser beam irradiation. Thus, grooves are formed near the conductor layers 18a to 18d in the upper principal surface of the insulator layer 16a and the lower principal surface of the insulator layer 16b. Formation of such grooves leads to a decrease in strength of the insulator layers 16a and 16b. In the circuit board 10, the first acute angle $\theta 1$ and the third acute angle $\theta 3$ are larger than the second acute angle $\theta 2$, the fourth acute angle $\theta 4$, the fifth acute angle $\theta 5$, the sixth acute angle $\theta 6$, the seventh acute angle $\theta 7$, and the eighth acute angle $\theta 8$. That is, the second side surface S2, the fourth side surface S4, the fifth side surface S5, the sixth side surface S6, the seventh side surface S7, and the eighth side surface S8 are formed by photolithography. The first side surface S1 and the third side surface S3 are formed by laser beam irradiation. In the photolithography step, no grooves are formed in the upper principal surface of the insulator layer 16a and the lower principal surface of the insulator layer 16b. Consequently, by combining laser beam irradiation and photolithography, a decrease in strength of the insulator layers 16a and 16b can be reduced or prevented.

FIG. 3 is a sectional view of a circuit board 300 of a comparative example. The circuit board 300 differs from the circuit board 10 in that the first acute angle $\theta 1$ and the third acute angle $\theta 3$ are equal or substantially equal to the second acute angle $\theta 2$, the fourth acute angle $\theta 4$, the fifth acute angle $\theta 5$, the sixth acute angle $\theta 6$, the seventh acute angle $\theta 7$, and the eighth acute angle $\theta 8$ and that the first groove H1 is not provided. The first acute angle $\theta 1$ and the third acute angle $\theta 3$ in the circuit board 10 are larger than the first acute angle $\theta 1$ and the third acute angle $\theta 3$ in the circuit board 300. Thus, in the circuit board 10, a distance between the conductor layer 18a and the conductor layer 18b can be narrower. When the first groove H1 is empty or when the first groove H1 is filled with a low dielectric constant material, an increase in a capacitance due to the narrow distance between the conductor layer 18a and the conductor layer 18b can be reduced or prevented.

In the circuit board 10, the first groove H1 is filled with the insulating material having a dielectric constant lower than that of the first insulator layer 116a. This reduces capacitance generated between the conductor layer 18a and the conductor layer 18b.

First Modification

Structure of Circuit Board 10a

A structure of a circuit board 10a according to a first modification of a preferred embodiment of the present invention will be described below. FIG. 4 is a top view of the

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circuit board **10a**. FIG. **5** is a sectional view of the circuit board **10a** taken along line B-B. In FIG. **4**, the protective layer **20a** is omitted.

The circuit board **10a** differs from the circuit board **10** in the following points.

The circuit board **10a** does not include the insulator layer **16b**, the conductor layers **18c** and **18d**, and the protective layer **20b**.

The protective layer **20a** partially covers both of the conductor layers **18a** and **18b**.

To be more specific, the substrate body **12** includes one insulator layer **16a**. The substrate body **12** is, for example, a laminated substrate made of a thermoplastic resin such as a liquid crystal polymer. Alternatively, for example, the substrate body **12** may be a laminated substrate made of a glass epoxy resin in which glass cloth is solidified with an epoxy resin. The multiple conductor layers **18a** and **18b** include the pair **P1** of the conductor layer **18a** (first conductor layer) and the conductor layer **18b** (second conductor layer) provided on the upper principal surface of the insulator layer **16a**. The circuit board **10a** includes only one pair of the first conductor layer and the second conductor layer provided on the insulator layer. Consequently, a pair of the first conductor layer and the second conductor layer having the smallest proximity distance is the pair **P1** of the conductor layer **18a** (first conductor layer) and the conductor layer **18b** (second conductor layer). Thus, the pair **P1** of the conductor layer **18a** (first conductor layer) and the conductor layer **18b** (second conductor layer) is defined as the pair **P11** of the most proximate first conductor layer **118a** and the most proximate second conductor layer **118b**.

The most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** are provided on the upper principal surface of the insulator layer **16a**. Thus, the insulator layer **16a** is defined as the first insulator layer **116a**. When viewed in the up-down direction, in at least a portion of an area located between the most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** in the first proximity section **A1** in the first insulator layer **116a**, the first groove **H1** is provided in the upper principal surface of the first insulator layer **116a**. The most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** are not physically connected.

The protective layer **20a** is provided on the upper principal surface of the first insulator layer **116a**. The protective layer **20a** covers the second side surface **S2** and the fourth side surface **S4** and does not cover the first side surface **S1** or the third side surface **S3**. Thus, the first groove **H1** is not filled with an insulating material. That is, the inside of the first groove **H1** is empty. Since other structures of the circuit board **10a** are the same or substantially the same as those of the circuit board **10**, description thereof will be omitted.

Method of Manufacturing Circuit Board **10a**

Next, a non-limiting example of a method of manufacturing the circuit board **10a** will be described with reference to the accompanying drawings. FIGS. **6**, **8**, and **10** are top views of the circuit board **10a** during manufacture. FIGS. **7**, **9**, and **11** are sectional views of the circuit board **10a** during manufacture. In an actual method of manufacturing the circuit board **10a**, multiple circuit boards **10a** are manufactured at the same time by cutting a large-sized first insulator layer **116a**. However, the method of manufacturing one circuit board **10a** will be described below.

As illustrated in FIGS. **6** and **7**, an insulator layer with a metal foil is prepared (preparing step) in which a metal foil **117** is provided on the upper principal surface of the first

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insulator layer **116a** including the upper principal surface and the lower principal surface aligned in the up-down direction. The metal foil **117** is, for example, a copper foil. The metal foil **117** covers the entire or substantially the entire upper principal surface of the first insulator layer **116a**.

After the preparing step, as illustrated in FIGS. **8** and **9**, photolithography is performed on the metal foil **117** to form a conductor layer **119** (pattern forming step). In the pattern forming step, a mask having an outer shape along the second side surface **S2** and the fourth side surface **S4** is formed on the metal foil **117**. Then, the metal foil **117** is etched using the mask. An etchant is, for example, an aqueous iron chloride solution. This forms the second side surface **S2** and the fourth side surface **S4**. However, in the pattern forming step, the first side surface **S1** and the third side surface **S3** are not formed. Consequently, the conductor layer **119** has a shape in which the conductor layer **18a** and the conductor layer **18b** are connected as one.

After the pattern forming step, as illustrated in FIGS. **10** and **11**, the conductor layer **119** is linearly irradiated with a laser beam when viewed in the up-down direction to separate the conductor layer **119** into the conductor layer **18a** (first conductor layer) and the conductor layer **18b** (second conductor layer) with the area linearly irradiated with the laser beam interposed therebetween (beam irradiation step). The laser beam is, for example, a UV laser beam. In the beam irradiation step, the first groove **H1** is formed in the upper principal surface of the first insulator layer **116a**. The lower end of the first groove **H1** does not extend to the lower principal surface of the first insulator layer **116a**. The lower end of the first groove **H1** is located above the lower principal surface of the first insulator layer **116a**.

After the beam irradiation step, as illustrated in FIGS. **4** and **5**, the protective layer **20a** is formed on the upper principal surface of the first insulator layer **116a**. Through the above steps, the circuit board **10a** is completed.

Advantageous Effects

According to the circuit board **10a**, the most proximate first conductor layer **118a** (conductor layer **18a**) and the most proximate second conductor layer **118b** (conductor layer **18b**), which are close to each other, can be formed with high accuracy for the same reason as for the circuit board **10**. According to the circuit board **10a**, the conductor layers **18a** and **18b** can be formed in a short time for the same reason as for the circuit board **10**. According to the circuit board **10a**, a decrease in strength of the insulator layer **16a** can be reduced or prevented for the same reason as for the circuit board **10**. According to the circuit board **10a**, capacitance generated between the conductor layer **18a** and the conductor layer **18b** can be reduced for the same reason as for the circuit board **10**.

In the circuit board **10a**, the protective layer **20a** covers the second side surface **S2** and the fourth side surface **S4** and does not cover the first side surface **S1** or the third side surface **S3**. Thus, the first groove **H1** is not filled with an insulating material. The inside of the first groove **H1** is empty. Consequently, an increase in the dielectric constant in the first groove **H1** can be reduced or prevented. This reduces the capacitance generated between the conductor layer **18a** and the conductor layer **18b**.

According to the above-described method of manufacturing the circuit board **10a**, in the beam irradiation step, the first groove **H1** is formed in the first insulator layer **116a**. The lower end of the first groove **H1** does not extend to the

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lower principal surface of the first insulator layer **116a**. This suppresses misalignment between the conductor layer **18a** and the conductor layer **18b**.

Second Modification

A structure of a circuit board **10b** according to a second modification of a preferred embodiment of the present invention will be described below. FIG. **12** is a top view of the circuit board **10b**. FIG. **13** is a sectional view of the circuit board **10b** taken along line B-B. In FIG. **12**, the protective layer **20a** is omitted.

The circuit board **10b** differs from the circuit board **10a** in that the circuit board **10b** has a coplanar structure. To be more specific, the circuit board **10b** further includes a most proximate third conductor layer **118c** (conductor layer **18c**). The most proximate third conductor layer **118c** is provided on the upper principal surface of the first insulator layer **116a**. The most proximate third conductor layer **118c** has a linear shape extending in the right-left direction.

The pair P11 of the most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** includes the most proximate section A11 in which the most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** are aligned in the back-forth direction (orthogonal direction). The most proximate first conductor layer **118a** is provided between the most proximate second conductor layer **118b** and the most proximate third conductor layer **118c** in the back-forth direction (orthogonal direction) in the most proximate section A11. That is, the most proximate second conductor layer **118b** is located in front of the most proximate first conductor layer **118a**. The most proximate third conductor layer **118c** is located behind the most proximate first conductor layer **118a**.

The most proximate second conductor layer **118b** and the most proximate third conductor layer **118c** are connected to ground potential. That is, the most proximate second conductor layer **118b** and the most proximate third conductor layer **118c** are ground conductor layers.

When viewed in the up-down direction, in an area located between the most proximate first conductor layer **118a** and the most proximate third conductor layer **118c** in the most proximate section A11 in the first insulator layer **116a**, a second groove H2 is provided in the upper principal surface of the first insulator layer **116a**. Since a shape of the second groove H2 is the same as the shape of the first groove H1, description thereof is omitted.

The protective layer **20a** is provided on the upper principal surface of the first insulator layer **116a**. The protective layer **20a** covers the fourth side surface S4 and the sixth side surface S6 and does not cover the first side surface S1, the second side surface S2, the third side surface S3, or the fifth side surface S5. Thus, the first groove H1 and the second groove H2 are not filled with an insulating material. That is, the inside of the first groove H1 and the inside of the second groove H2 are empty. Since other structures of the circuit board **10b** are the same as those of the circuit board **10a**, description thereof will be omitted.

According to the circuit board **10b**, the most proximate first conductor layer **118a** (conductor layer **18a**), the most proximate second conductor layer **118b** (conductor layer **18b**), and the most proximate third conductor layer **118c** (conductor layer **18c**), which are close to each other, can be provided with high accuracy for the same reason as for the circuit board **10a**. According to the circuit board **10b**, the most proximate first conductor layer **118a**, the most proximate

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second conductor layer **118b**, and the most proximate third conductor layer **118c** can be formed in a short time for the same reason as for the circuit board **10a**. According to the circuit board **10b**, a decrease in strength of the insulator layer **16a** can be reduced or prevented for the same reason as for the circuit board **10a**. According to the circuit board **10b**, capacitance generated between the most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** and capacitance generated between the most proximate first conductor layer **118a** and the most proximate third conductor layer **118c** can be reduced for the same reason as for the circuit board **10a**.

Third Modification

A structure of a circuit board **10c** according to a third modification of a preferred embodiment of the present invention will be described below. FIG. **14** is a top view of the circuit board **10c**. FIG. **15** is a sectional view of the circuit board **10c** taken along line E-E. In FIG. **14**, the protective layer **20a** is omitted.

The circuit board **10c** differs from the circuit board **10a** in that the conductor layers **18a** to **18d** define a coil C. To be more specific, the circuit board **10c** includes the conductor layers **18a** to **18d**. The conductor layers **18a** to **18d** are provided on the upper principal surface of the first insulator layer **116a**. The conductor layers **18a** to **18d** have linear shapes extending in the back-forth direction. The conductor layer **18d**, the conductor layer **18b**, the conductor layer **18a**, and the conductor layer **18c** are aligned in this order from left to right. The conductor layers **18a** to **18d** are electrically connected. To be more specific, the conductor layer **18d**, the conductor layer **18b**, the conductor layer **18a**, and the conductor layer **18c** are electrically connected in series in this order to define the coil C having a spiral shape when viewed in the up-down direction.

Here, the conductor layer **18a** is the most proximate first conductor layer **118a**. The conductor layer **18b** is the most proximate second conductor layer **118b**. The conductor layer **18c** is the most proximate third conductor layer **118c**. The conductor layer **18d** is a most proximate fourth conductor layer **118d**. When viewed in the up-down direction, in at least a portion of an area located between the most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** in the first proximity section A1 in the first insulator layer **116a**, the first groove H1 is provided in the upper principal surface of the first insulator layer **116a**. When viewed in the up-down direction, in at least a portion of an area located between the most proximate first conductor layer **118a** and the most proximate third conductor layer **118c** in the first proximity section A2 in the first insulator layer **116a**, the first groove H1 is provided in the upper principal surface of the first insulator layer **116a**. When viewed in the up-down direction, in at least a portion of an area located between the most proximate second conductor layer **118b** and the most proximate fourth conductor layer **118d** in a first proximity section A3 in the first insulator layer **116a**, the first groove H1 is provided in the upper principal surface of the first insulator layer **116a**. Further, the three first grooves H1 extending in the front-back direction are connected by the two first grooves H1 extending in the front-back direction. Thus, when viewed in the up-down direction, the first grooves H1 define a spiral shape along the most proximate first conductor layer **118a**, the most proximate second conductor layer **118b**, the most proximate third conductor layer **118c**, and the most proximate fourth conductor layer **118d**. As described above, in the

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circuit board 10c, the spiral shape is provided by the first grooves H1 in a rectangular or substantially rectangular conductor layer, thus providing the spiral-shaped coil C. Since other structures of the circuit board 10c are the same or substantially the same as those of the circuit board 10a, description thereof will be omitted.

According to the circuit board 10c, the most proximate first conductor layer 118a, the most proximate second conductor layer 118b, the most proximate third conductor layer 118c, and the most proximate fourth conductor layer 118d, which are close to each other, can be provided with high accuracy for the same reason as for the circuit board 10a. According to the circuit board 10c, the most proximate first conductor layer 118a, the most proximate second conductor layer 118b, the most proximate third conductor layer 118c, and the most proximate fourth conductor layer 118d can be formed in a short time for the same reason as for the circuit board 10a. According to the circuit board 10c, a decrease in strength of the insulator layer 16a can be reduced or prevented for the same reason as for the circuit board 10a. According to the circuit board 10c, capacitance generated between the most proximate first conductor layer 118a and the most proximate second conductor layer 118b, capacitance generated between the most proximate first conductor layer 118a and the most proximate third conductor layer 118c, and capacitance generated between the most proximate second conductor layer 118b and the most proximate fourth conductor layer 118d can be reduced for the same reason as for the circuit board 10a. Further, according to the circuit board 10c, the coil C including a large number of turns can be provided in a small space. Thus, the coil C having a high Q factor can be obtained.

Fourth Modification

A structure of a circuit board 10d according to a fourth modification of a preferred embodiment of the present invention will be described below. FIG. 16 is an exploded perspective view of the circuit board 10d. FIG. 17 is a sectional view of the circuit board 10d taken along line C-C.

The circuit board 10d differs from the circuit board 10 in that the circuit board 10d has a microstripline structure. To be more specific, the circuit board 10d includes a ground conductor layer 50, instead of the conductor layers 18c and 18d. The ground conductor layer 50 covers or substantially covers the entire lower principal surface of the insulator layer 16b. Thus, when viewed in the up-down direction, the conductor layers 18a and 18b overlap the ground conductor layer 50. That is, the conductor layers 18a and 18b and the ground conductor layer 50 define the microstripline structure.

The conductor layer 18a is the most proximate first conductor layer 118a. The conductor layer 18b is the most proximate second conductor layer 118b. The pair P11 of the most proximate first conductor layer 118a and the most proximate second conductor layer 118b include the most proximate section A11 and a second proximity section A20 in which the most proximate first conductor layer 118a and the most proximate second conductor layer 118b are aligned in the back-forth direction (orthogonal direction). The second proximity section A20 is located to the left of the most proximate section A11. A distance D200 between the most proximate first conductor layer 118a and the most proximate second conductor layer 118b in the second proximity section A20 is greater than the first most proximate inter-conductor distance D11.

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When viewed in the up-down direction, in an area located between the most proximate first conductor layer 118a and the most proximate second conductor layer 118b in the second proximity section A20 in the first insulator layer 116a, no groove is provided in the upper principal surface of the first insulator layer 116a.

The circuit board 10d also includes conductor layers 30a and 30b. The conductor layers 30a and 30b are provided on the upper principal surface of the first insulator layer 116a. To be more specific, the conductor layer 30a is located behind the most proximate first conductor layer 118a in the most proximate section A11. The conductor layer 30b is located in front of the most proximate second conductor layer 118b in the most proximate section A11.

The circuit board 10d further includes a first outer electrode 60a and a second outer electrode 60b. The first outer electrode 60a and the second outer electrode 60b are provided on the substrate body 12. In the present modification, the first outer electrode 60a and the second outer electrode 60b are provided on the upper principal surface of the first insulator layer 116a. The first outer electrode 60a is electrically connected to the most proximate first conductor layer 118a. The first outer electrode 60a is connected to a left end of the most proximate first conductor layer 118a. The second outer electrode 60b is electrically connected to the most proximate second conductor layer 118b. The second outer electrode 60b is connected to a left end of the most proximate second conductor layer 118b.

The protective layer 20a includes openings h1 and h2. The first outer electrode 60a and the second outer electrode 60b are exposed to the outside of the circuit board 10d from the openings h1 and h2, respectively. Thus, connectors and electronic components can be mounted on the first outer electrode 60a and the second outer electrode 60b. Since other structures of the circuit board 10d are the same or substantially the same as those of the circuit board 10, description thereof will be omitted.

According to the circuit board 10d, the most proximate first conductor layer 118a (conductor layer 18a) and the most proximate second conductor layer 118b (conductor layer 18b), which are close to each other, can be provided with high accuracy for the same reason as for the circuit board 10. According to the circuit board 10d, the most proximate first conductor layer 118a and the most proximate second conductor layer 118b can be formed in a short time for the same reason as for the circuit board 10. According to the circuit board 10d, a decrease in strength of the insulator layer 16a can be reduced or prevented for the same reason as for the circuit board 10. According to the circuit board 10d, capacitance generated between the most proximate first conductor layer 118a and the most proximate second conductor layer 118b can be reduced for the same reason as for the circuit board 10.

Including the conductor layers 30a and 30b, as in the circuit board 10d, requires a small distance between the most proximate first conductor layer 118a and the most proximate second conductor layer 118b. In the circuit board 10d, even when the distance between the most proximate first conductor layer 118a and the most proximate second conductor layer 118b is reduced, the capacitance generated between the most proximate first conductor layer 118a and the most proximate second conductor layer 118b can be reduced. Further, even when the distance between the most proximate first conductor layer 118a and the most proximate second conductor layer 118b is reduced, the first groove H1 reduces or prevents electric field coupling between the most proximate first conductor layer 118a and the most proximate

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second conductor layer **118b**. As a result, a characteristic impedance of the circuit board **10d** is less likely to change from a desired characteristic impedance (e.g., about 50Ω). When viewed in the up-down direction, in the area located between the most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** in the second proximity section **A20** in the first insulator layer **116a**, a groove may be provided in the upper principal surface of the first insulator layer **116a**. Electronic components may be provided instead of the conductor layers **30a** and **30b**.

Fifth Modification

A structure of a circuit board **10e** according to a fifth modification of a preferred embodiment of the present invention will be described below. FIG. **18** is a sectional view of the circuit board **10e**.

The circuit board **10e** differs from the circuit board **10d** in that the circuit board **10e** further includes an adhesive layer **90**. The most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** are fixed to the upper principal surface of the first insulator layer **116a** with the adhesive layer **90** interposed therebetween. The first groove **H1** passes through the adhesive layer **90** in the up-down direction. A dielectric constant of the adhesive layer **90** is lower than the dielectric constant of the first insulator layer **116a**. Thus, capacitance formed between the most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** can be reduced.

The most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** are provided on the adhesive layer **90**, and the first insulator layer **116a** is provided under the adhesive layer **90**. The dielectric constant of the first insulator layer **116a** is higher than the dielectric constant of the adhesive layer **90**. This reduces or prevents spread of the electric field generated by the most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** around the circuit board **10e**. Since other structures of the circuit board **10e** are the same as those of the circuit board **10d**, description thereof will be omitted.

In a non-limiting example of a method of manufacturing the circuit board **10e**, the first groove **H1** is formed in the first insulator layer **116a** and the adhesive layer **90** in the beam irradiation step. The lower end of the first groove **H1** does not extend to the lower principal surface of the first insulator layer **116a**. After the beam irradiation step, the insulator layer **16b** is stacked on the first insulator layer **116a**. This reduces or prevents misalignment between the conductor layer **18a** and the conductor layer **18b**.

According to the circuit board **10e**, the most proximate first conductor layer **118a** (conductor layer **18a**) and the most proximate second conductor layer **118b** (conductor layer **18b**), which are close to each other, can be provided with high accuracy for the same reason as for the circuit board **10d**. According to the circuit board **10e**, the most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** can be formed in a short time for the same reason as for the circuit board **10d**. According to the circuit board **10e**, a decrease in strength of the insulator layer **16a** can be reduced or prevented for the same reason as for the circuit board **10d**. According to the circuit board **10e**, the capacitance generated between the most proximate first conductor layer **118a** and the most proximate

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second conductor layer **118b** can be reduced for the same reason as for the circuit board **10d**.

Sixth Modification

A structure of a circuit board **10f** according to a sixth modification of a preferred embodiment of the present invention will be described below. FIG. **19** is an exploded perspective view of the circuit board **10f**. FIG. **20** is a sectional view of the circuit board **10f** taken along line C-C.

The circuit board **10f** differs from the circuit board **10d** in that the circuit board **10f** has a stripline structure. The substrate body **12** further includes the second insulator layer **116b**. The second insulator layer **116b** is located on the first insulator layer **116a**. The second insulator layer **116b** is stacked on the first insulator layer **116a**. The material of the first insulator layer **116a** is the same as the material of the second insulator layer **116b**. A portion of the second insulator layer **116b** is located in the first groove **H1** formed in the first insulator layer **116a**. However, the first groove **H1** is formed by, for example, laser beam irradiation. Consequently, a molecular structure of the inner surface of the first groove **H1** is slightly changed by heating. Consequently, the inner surface of the first groove **H1** can be confirmed with an electronic microscope such as an SEM, for example.

The circuit board **10f** further includes a ground conductor layer **52** and multiple via-hole conductors **v1** and **v2**. The ground conductor layer **52** covers or substantially covers an entire upper principal surface of the second insulator layer **116b**. Thus, the ground conductor layer **52** overlaps the most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** when viewed in the up-down direction.

The ground conductor layer **50** covers or substantially covers the entire lower principal surface of the first insulator layer **116a**. Thus, the ground conductor layer **50** overlaps the most proximate first conductor layer **118a** and the most proximate second conductor layer **118b** when viewed in the up-down direction. Thus, the most proximate first conductor layer **118a**, the most proximate second conductor layer **118b**, and the ground conductor layers **50** and **52** define a stripline structure.

The multiple via-hole conductors **v1** and **v2** electrically connect the ground conductor layer **50** and the ground conductor layer **52**. The multiple via-hole conductors **v1** and **v2** pass through the first insulator layer **116a** and the second insulator layer **116b** in the up-down direction. The multiple via-hole conductors **v1** are aligned in a line in the right-left direction behind the most proximate first conductor layer **118a** and the most proximate second conductor layer **118b**. The multiple via-hole conductors **v2** are aligned in a line in the right-left direction in front of the most proximate first conductor layer **118a** and the most proximate second conductor layer **118b**.

The first outer electrode **60a** and the second outer electrode **60b** are provided on the upper principal surface of the first insulator layer **116a**. The first outer electrode **60a** overlaps the left end portion of the most proximate first conductor layer **118a** when viewed in the up-down direction. The second outer electrode **60b** overlaps the left end portion of the most proximate second conductor layer **118b** when viewed in the up-down direction.

The circuit board **10f** further includes via-hole conductors **v3** and **v4**. The via-hole conductor **v3** electrically connects the first outer electrode **60a** to the left end portion of the most proximate first conductor layer **118a**. The via-hole conductor **v4** electrically connects the second outer elec-

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trode 60b to the left end portion of the most proximate second conductor layer 118b. The via-hole conductors v3 and v4 pass through the second insulator layer 116b in the up-down direction.

The protective layer 20a includes openings h1 to h4. The first outer electrode 60a and the second outer electrode 60b are exposed to the outside of the circuit board 10f from the openings h1 and h2, respectively. A portion of the ground conductor layer 52 is exposed from the openings h3 and h4 to the outside of the circuit board 10f. Thus, connectors and electronic components can be mounted on the first outer electrode 60a, the second outer electrode 60b, and the ground conductor layer 52. Since other structures of the circuit board 10f are the same or substantially the same as those of the circuit board 10d, description thereof will be omitted.

According to the circuit board 10f, the most proximate first conductor layer 118a (conductor layer 18a) and the most proximate second conductor layer 118b (conductor layer 18b), which are close to each other, can be provided with high accuracy for the same reason as for the circuit board 10d. According to the circuit board 10f, the most proximate first conductor layer 118a and the most proximate second conductor layer 118b can be formed in a short time for the same reason as for the circuit board 10d. According to the circuit board 10f, a decrease in strength of the insulator layer 16a can be reduced or prevented for the same reason as for the circuit board 10d.

Seventh Modification

A structure of a circuit board 10g according to a seventh modification of a preferred embodiment of the present invention will be described below. FIG. 21 is a sectional view of the circuit board 10g.

The circuit board 10g differs from the circuit board 10f in that the circuit board 10g further includes a second adhesive layer 92. The second insulator layer 116b is fixed to the first insulator layer 116a with the second adhesive layer 92 interposed therebetween. A dielectric constant of the second adhesive layer 92 is lower than the dielectric constant of the first insulator layer 116a. A portion of the second adhesive layer 92 is located in the first groove H1. However, the portion of the second adhesive layer 92 does not have to be located in the first groove H1. Since other structures of the circuit board 10g are the same as those of the circuit board 10f, description thereof will be omitted. Note that the first groove H1 does not have to be completely filled with the second adhesive layer 92.

According to the circuit board 10g, the most proximate first conductor layer 118a (conductor layer 18a) and the most proximate second conductor layer 118b (conductor layer 18b), which are close to each other, can be provided with high accuracy for the same reason as for the circuit board 10f. According to the circuit board 10g, the most proximate first conductor layer 118a and the most proximate second conductor layer 118b can be formed in a short time for the same reason as for the circuit board 10f. According to the circuit board 10g, a decrease in strength of the insulator layer 16a can be reduced or prevented for the same reason as for the circuit board 10f. According to the circuit board 10g, capacitance generated between the most proximate first conductor layer 118a and the most proximate second conductor layer 118b can be reduced for the same reason as for the circuit board 10f.

Other Preferred Embodiments

The circuit boards according to preferred embodiments of the present invention is not limited to the circuit boards 10

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and 10a to 10g and can be changed within the scope of the gist thereof. In addition, the structures of the circuit boards 10 and 10a to 10g may be combined as desired.

In the circuit boards 10 and 10a to 10g, the most proximate second conductor layer 118b does not have to be a signal line. The most proximate second conductor layer 118b may be, for example, a ground conductor layer or a power supply line.

In the circuit boards 10 and 10a to 10g, the protective layers 20a and 20b are not necessary.

In the circuit boards 10 and 10a to 10g, the maximum value D100 of the width of the first groove H1 in the back-forth direction (orthogonal direction) may be shorter than the first most proximate inter-conductor distance D11. In this case, the upper principal surface of the first insulator layer 116a is visible between the first side surface S1 and the inner surface of the first groove H1. The upper principal surface of the first insulator layer 116a is visible between the third side surface S3 and the inner surface of the first groove H1.

In the circuit boards 10 and 10a to 10g, when the first groove H1 is formed by laser beam irradiation, for example, the first groove H1 may become wider in some cases. In this case, the first groove H1 may extend under the most proximate first conductor layer 118a and under the most proximate second conductor layer 118b.

In the circuit boards 10 and 10d to 10f, the lower end of the first groove H1 may extend below the lower principal surface of the first insulator layer 116a. In this case, the laser beam irradiation is performed after stacking the first insulator layer 116a and the insulator layer 16b.

The first groove H1 may extend to a lower end of the first insulator layer 116a.

The surface roughness of the inner surface s0 of the first groove H1 may be larger or smaller than the surface roughness of the surface s11 of the first insulator layer 116a with which the most proximate first conductor layer 118a is in contact.

While preferred embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

What is claimed is:

1. A circuit board comprising:

a substrate body including one or more insulator layers, an upper principal surface, and a lower principal surface with a normal line extending in an up-down direction; and

a plurality of conductor layers in or on the substrate body; wherein

the plurality of conductor layers include one or more pairs of a first conductor layer and a second conductor layer provided on one of the one or more insulator layers; each of the one or more pairs of the first conductor layer and the second conductor layer includes a first proximity section in which the first conductor layer and the second conductor layer are aligned in an orthogonal direction orthogonal or substantially orthogonal to an extending direction of the first conductor layer;

a distance between the first conductor layer and the second conductor layer in the first proximity section is defined as a proximity distance;

the one or more insulator layers includes a first insulator layer;

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a most proximate first conductor layer and a most proximate second conductor layer are defined as the first and second conductor layers with a smallest proximity distance and are located on an upper principal surface of the first insulator layer;

the most proximate first conductor layer includes a first side surface and a second side surface aligned in the orthogonal direction;

the most proximate second conductor layer includes a third side surface and a fourth side surface aligned in the orthogonal direction;

the first side surface and the third side surface face each other; and

a first acute angle between the first side surface and the upper principal surface of the first insulator layer is larger than a second acute angle between the second side surface and the upper principal surface of the first insulator layer.

2. The circuit board according to claim 1, wherein the first conductor layer has a linear shape when viewed in the up-down direction;

a pair of the first conductor layer and the second conductor layer having a smallest proximity distance is defined as a pair of the most proximate first conductor layer and the most proximate second conductor layer; and

when viewed in the up-down direction, in at least a portion of an area located between the most proximate first conductor layer and the most proximate second conductor layer in the first proximity section in the first insulator layer, a first groove is provided in the upper principal surface of the first insulator layer.

3. The circuit board according to claim 2, wherein the most proximate first conductor layer includes a first side surface and a second side surface aligned in the orthogonal direction;

the most proximate second conductor layer includes a third side surface and a fourth side surface aligned in the orthogonal direction;

the first side surface and the third side surface face each other;

the upper principal surface of the first insulator layer is not visible between the first side surface and an inner surface of the first groove; and

the upper principal surface of the first insulator layer is not visible between the third side surface and the inner surface of the first groove.

4. The circuit board according to claim 2, wherein the proximity distance between the most proximate first conductor layer and the most proximate second conductor layer in the first proximity section is defined as a first most proximate inter-conductor distance, and a maximum value of a width of the first groove in the orthogonal direction is greater than or equal to the first most proximate inter-conductor distance.

5. The circuit board according to claim 2, wherein a lower end of the first groove is located above a lower principal surface of the first insulator layer.

6. The circuit board according to claim 2, wherein the most proximate first conductor layer and the most proximate second conductor layer are fixed to the upper principal surface of the first insulator layer with an adhesive layer interposed between the most proximate first conductor layer and the most proximate second conductor layer and the first insulator layer; and the first groove passes through the adhesive layer in the up-down direction.

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7. The circuit board according to claim 2, wherein the first groove is filled with an insulating material having a dielectric constant lower than a dielectric constant of the first insulator layer.

8. The circuit board according to claim 1, wherein the first conductor layer is a signal line to transmit a first high-frequency signal.

9. The circuit board according to claim 1, wherein the most proximate second conductor layer is a signal line to transmit a second high-frequency signal.

10. The circuit board according to claim 9, wherein the most proximate first conductor layer and the most proximate second conductor layer include a most proximate section and a second proximity section in which the most proximate first conductor layer and the most proximate second conductor layer are aligned in the orthogonal direction;

a distance between the most proximate first conductor layer and the most proximate second conductor layer in the most proximate section is defined as a first most proximate inter-conductor distance;

a distance between the most proximate first conductor layer and the most proximate second conductor layer in the second proximity section is greater than the first most proximate inter-conductor distance; and

when viewed in the up-down direction, in an area located between the most proximate first conductor layer and the most proximate second conductor layer in the second proximity section in the first insulator layer, no groove is provided in the upper principal surface of the first insulator layer.

11. The circuit board according to claim 9, further comprising:

a first outer electrode provided on the substrate body and electrically connected to the most proximate first conductor layer; and

a second outer electrode provided on the substrate body and electrically connected to the most proximate second conductor layer.

12. The circuit board according to claim 1, further comprising:

a most proximate third conductor layer provided on the upper principal surface of the first insulator layer; wherein the most proximate first conductor layer and the most proximate second conductor layer include a most proximate section in which the most proximate first conductor layer and the most proximate second conductor layer are aligned in the orthogonal direction;

the most proximate first conductor layer is provided between the most proximate second conductor layer and the most proximate third conductor layer in the orthogonal direction in the most proximate section;

the most proximate second conductor layer and the most proximate third conductor layer are connected to a ground potential; and

when viewed in the up-down direction, in an area located between the most proximate first conductor layer and the most proximate third conductor layer in the most proximate section in the first insulator layer, a second groove is provided in the upper principal surface of the first insulator layer.

13. The circuit board according to claim 2, wherein the one or more insulator layers further includes a second insulator layer located on the first insulator layer;

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the second insulator layer is fixed to the first insulator layer with a second adhesive layer interposed between the first insulator layer and the second insulator layer; a dielectric constant of the second adhesive layer is lower than a dielectric constant of the first insulator layer; and a portion of the second adhesive layer is located in the first groove.

14. The circuit board according to claim **1**, wherein

the most proximate first conductor layer includes a first side surface and a second side surface aligned in the orthogonal direction;

the most proximate second conductor layer includes a third side surface and a fourth side surface aligned in the orthogonal direction;

the first side surface and the third side surface face each other; and

the circuit board further comprises:

a protective layer provided on the upper principal surface of the first insulator layer, covering the second side surface and the fourth side surface, and not covering the first side surface or the third side surface.

15. A circuit board comprising:

a substrate body including one or more insulator layers, an upper principal surface, and a lower principal surface with a normal line extending in an up-down direction; and

a plurality of conductor layers in or on the substrate body; wherein

the plurality of conductor layers include one or more pairs of a first conductor layer and a second conductor layer provided on one of the one or more insulator layers;

each of the one or more pairs of the first conductor layer and the second conductor layer includes a first proximity section in which the first conductor layer and the second conductor layer are aligned in an orthogonal direction orthogonal or substantially orthogonal to an extending direction of the first conductor layer;

a distance between the first conductor layer and the second conductor layer in the first proximity section is defined as a proximity distance;

the one or more insulator layers includes a first insulator layer;

a most proximate first conductor layer and a most proximate second conductor layer are provided on an upper principal surface of the first insulator layer;

a groove is provided between the first and second conductor layers; and

no groove is provided on either of outer sides of the first and second conductor layers.

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16. The circuit board according to claim **15**, wherein the most proximate first conductor layer includes a first side surface and a second side surface aligned in the orthogonal direction;

the most proximate second conductor layer includes a third side surface and a fourth side surface aligned in the orthogonal direction;

the first side surface and the third side surface face each other; and

a first acute angle between the first side surface and the upper principal surface of the first insulator layer is larger than a second acute angle between the second side surface and the upper principal surface of the first insulator layer.

17. The circuit board according to claim **15**, wherein the first conductor layer is a signal line to transmit a first high-frequency signal.

18. The circuit board according to claim **15**, wherein the most proximate second conductor layer is a signal line to transmit a second high-frequency signal.

19. The circuit board according to claim **18**, wherein the most proximate first conductor layer and the most proximate second conductor layer include a most proximate section and a second proximity section in which the most proximate first conductor layer and the most proximate second conductor layer are aligned in the orthogonal direction;

a distance between the most proximate first conductor layer and the most proximate second conductor layer in the most proximate section is defined as a first most proximate inter-conductor distance;

a distance between the most proximate first conductor layer and the most proximate second conductor layer in the second proximity section is greater than the first most proximate inter-conductor distance; and

when viewed in the up-down direction, in an area located between the most proximate first conductor layer and the most proximate second conductor layer in the second proximity section in the first insulator layer, no groove is provided in the upper principal surface of the first insulator layer.

20. The circuit board according to claim **18**, further comprising:

a first outer electrode provided on the substrate body and electrically connected to the most proximate first conductor layer; and

a second outer electrode provided on the substrate body and electrically connected to the most proximate second conductor layer.

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